

Exact test problems and a work–accuracy protocol for benchmarking nearest correlation matrix solvers

Oyeyinka E. Ibrahim
Department of Industrial and Systems Engineering
Northern Illinois University
ioyeyinka1@niu.edu

Abstract

Nearest correlation matrix (NCM) solvers are usually compared on matrices whose solutions are unknown, so the accuracy each delivers goes unmeasured. We specialize the known technique of generating test problems from prescribed optimality conditions to the NCM problem, obtaining instances with closed-form primal and dual solutions and prescribed rank, multiplicity and eigenvalue separation. The semismooth Newton method with the Borsdorf–Higham line search is cheapest under all 11 conventions examined, with a median eigendecomposition count, at the tightest target, of four or five on every tested exact-solution family, from $n = 100$ to $n = 2105$. Second at $n = 100$, under ten of the eleven conventions and tied under the other, is Anderson-accelerated alternating projections. The ordering of the remaining methods is a property of the test set, changing above all with the solution rank: a published method, and our line-search adaptation of another method, exchange places between $r = 20$ and $r = 50$ at $n = 100$. Under the dimension-scaled rule $\|\nabla\theta\|_2 \leq 10^{-7}n$, first-order methods return, in median on 18 paired instances, about ten times less accuracy at $n = 500$ than at $n = 100$, so a larger problem can appear easier. A known finite-precision failure of objective-value Armijo tests recurs in a second published method, where our adaptation of a safeguard from the NCM literature removes it.

Keywords. nearest correlation matrix; semidefinite programming; test problem generation; benchmarking; semismooth Newton method; reproducibility.

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1 Introduction

Given a symmetric matrix $G \in \mathbb{R}^{n \times n}$, the nearest correlation matrix problem asks for

$$\min_X \frac{1}{2} \|X - G\|_F^2 \quad \text{subject to} \quad X \succeq 0, \text{diag}(X) = e. \quad (1)$$

The problem has a unique solution. Since Higham’s alternating-projections method [18], the literature has produced dual formulations and first-order schemes [27, 8], a quadratically convergent semismooth Newton method [31], its preconditioned and globalized refinement [6], accelerated projection methods [20], and, most recently, accelerated gradient methods with quasi-Newton preconditioning [22]. Most of these papers include numerical comparisons, and those comparisons are how a reader decides which method to use.

This paper is about how such comparisons are made. Its scope is narrow: to determine which methodological choices, when NCM solvers are compared, change the conclusion and which only change the numbers. Four candidate confounds are examined:

1. *Native stopping rules.* Each method stops when its own criterion is met, so the solvers stop at different accuracies.
2. *Approximate references.* Forward error is measured against the output of another solver instead of against the true solution.
3. *Rejected-iterate accounting.* Line-search trials that are rejected can be credited with reaching a target that the method has not actually reached.
4. *Hidden primitive work.* ‘Iterations’ count different operations in different methods, and some per-iteration work is not recorded at all.

A careful benchmark must control each of these. Without an exact solution, however, the second confound cannot be removed, and without trajectories the first and third cannot be separated.

Contributions. In brief, we give an exact-solution NCM family and a matched-accuracy protocol, and use them to show that conclusions drawn from NCM solver comparisons do not transfer across benchmark regimes. The rank at which an ordering established at $n = 100$ reverses moves higher at $n = 500$; the order also changes with the degree of invalidity on matrices built from equity returns, and of four matrices from the literature, the three that discriminate between the first-order methods are consistent with this dependence. We also show that a known finite-precision line-search defect recurs in a second published method and occurs on real matrices. Each contribution builds on established practice (section 1.1).

1. *An exact-solution NCM family* (section 2). We specialize the standard construction of test problems from prescribed optimality conditions [25, 9] to the NCM problem. For semidefinite programs, the same idea is due to Wei and Wolkowicz [33]. The parts specific to NCM are the projection identity $\Pi_{\mathcal{S}_+^n}(G + \text{Diag } y^*) = X^*$, independent control of the rank, the multiplicity of the nonpositive eigenvalue of $C(y^*)$ nearest zero and its distance from zero, a paired-seed design, sufficient conditions for $|G_{ij}| \leq 1$ together with a fixed-rank saturation result for $\max_{i \neq j} |X_{ij}^*|$, and released instance sets at $n = 100$ and $n = 500$. The strict-complementarity defect equals that multiplicity when the distance is zero and vanishes otherwise, so 45 of the 270 instances at each dimension are not strictly complementary.
2. *A matched-accuracy protocol for NCM* (section 3). We adapt data profiles [29], fixed-target evaluation [17] and work-precision diagrams [16] to spectral solvers. The unit of cost is one EVD, which we check against wall-clock time. Because the reference is exact, two rules become enforceable: rejected trials receive no accuracy credit, and accuracy is matched in forward error. Primitive work is reported separately.
3. *An invariant first place above a regime-dependent ordering* (section 5). Six variants of five methods, the published ones verified against their authors’ own results, were run on 270 instances at $n = 100$ and 270 at $n = 500$; all or most of them were also run on 135 further instances at higher rank, 125 built from two equity panels at $n = 550$ and $n = 2105$, and four matrices from the literature (section 5 states which). The cheapest method is the same under all 11 conventions examined, and its median eigendecomposition count, at the tightest

target, is four or five on every tested exact-solution family, from $n = 100$ to $n = 2105$; wall-clock time is not flat, since each eigendecomposition itself costs more at larger n . Second at $n = 100$ is Anderson-accelerated alternating projections [20] (Anderson-APM), under 10 of the 11 conventions and tied for second under the eleventh; it is second on most of the other exact-solution sets too, but not all (it is fourth at 10^{-2} at $n = 500$, and behind both AGD variants at $r = 100$ and $r = 200$ at $n = 2105$), and not on perturbed or literature matrices. Below the first place the ordering changes with the solution rank, with how far the input is from a correlation matrix, and with the accuracy target. The rank dependence holds without SBB-Dual, the author’s own method (see below): Dykstra-APM (alternating projections with Dykstra’s correction [18]) and AGD-SDAJ-BH (our line-search adaptation of the accelerated method of Huynh and Hwang [22]) exchange places between $r = 20$ and $r = 50$ at $n = 100$, and the exchange has not occurred by $r = 150$ at $n = 500$. At $n = 2105$ the order of Anderson-APM and AGD-SDAJ-BH is not monotone in rank: Anderson-APM is cheaper at $r = 50$ and $r = 400$, AGD-SDAJ-BH at $r = 100$ and $r = 200$. The same rank pattern appears between SBB-Dual and AGD-SDAJ-BH, whose exchange is bracketed between $r = 5$ and $r = 20$ at $n = 100$ and between $r = 50$ and $r = 75$ at $n = 500$. Only a family that varies the spectral structure at the solution in controlled steps makes such an association directly observable.

4. *Replication of a known line-search defect* (section 4). The failure of objective-value sufficient-decrease tests near a minimizer is well known [15]. For the NCM dual, Borsdorf analysed it [5, §4.7.1], and this led to the modified step selection of Borsdorf and Higham [6]. The new content is limited to two points. First, the defect recurs in a separately published first-order method [22]. Second, we quantify it on the controlled family: 56/270 stalls for semismooth Newton, removed by the Borsdorf–Higham globalization, and 18/270 for AGD-SDAJ, removed by our adaptation of its safeguard. We also document a second, simpler trap (a redundant Jacobian factorization).
5. *An NCM quantification of the stopping-rule confound* (section 6). Dolan, Moré and Munson [12] warned that native stopping tests confound comparisons. For NCM, the widely used rule $\|\nabla\theta\|_2 \leq 10^{-7}n$ (applied by the two projection methods to the diagonal of their positive semidefinite iterate) leaves the achieved forward error spread over a factor of 383 across solvers at $n = 100$, and it degrades accuracy with dimension unequally across methods.

What we do not claim. We do not claim that any method is best in general. The ranking results are for a synthetic family at $n = 100$ and $n = 500$, with instances built from equity returns at $n = 550$ and $n = 2105$ (sections 5.8 and 5.9) and four matrices from the literature (section 5.10) as external checks, all on one machine. SBB-Dual is a new method from the author’s companion manuscript in preparation [23]. It is specified in full in section E, receives the same exposition as the others, and its convergence theory is not used here. Without SBB-Dual, Newton-SIN-BH is still first, Anderson-APM is still second at $n = 100$, and each kind of dependence is still present for a pair that excludes it: on the rank, between Dykstra-APM and AGD-SDAJ-BH (section 5.5); on the accuracy target, between Anderson-APM and AGD-SDAJ-BH at $n = 500$ (section 5.6); and on the degree of invalidity, between Anderson-APM and AGD-SDAJ-BH on the equity-derived matrices (section 5.8). Two findings rest on SBB-Dual against AGD-SDAJ-BH specifically: the bracketed positions of the rank crossover at $n = 100$ and $n = 500$ (sections 5.5 and 5.7), and the claim that the rank-dependent ordering found at $n = 100$ recurs at a higher rank at $n = 500$ (section 8). For the one pair without SBB-Dual checked for that recurrence, Dykstra-APM against AGD-SDAJ-BH,

the exchange has not occurred by $r = 150$ at $n = 500$.

1.1 Related work

Benchmarking methodology. Performance profiles [11] compare solvers by the ratio of their cost to the best cost on each problem. Cost is measured at each solver’s own termination, so the solver’s stopping test implicitly sets the accuracy. Dolan, Moré and Munson [12] showed that differing stopping criteria bias such profiles, and proposed rerunning under a common optimality measure. Data profiles [29] count cost in a problem-independent unit and show how rankings change with the required accuracy. COCO [17] uses a ladder of fixed targets against known optima. Work–precision diagrams, standard in the comparison of ODE solvers [16], plot error against a reference solution versus work. Gould and Scott [14] show that performance-profile rankings below the leading solver can change when solvers are added or removed. Beiranvand, Hare and Lucet [4] survey best practice, including known-solution test problems and reproducibility. Our protocol follows these principles; our contribution is to make their exact-reference versions possible for the NCM problem. The reversals below second place in section 5 arise with a *fixed* solver set, from the accuracy target and the instance regime, so the mechanism differs from the one studied by Gould and Scott.

Test problems with known solutions. Generating test problems from prescribed optimality conditions is standard for unconstrained optimization [28] and for quadratic programming [25, 9]. Wei and Wolkowicz [33] generate semidefinite programs from a chosen optimal primal–dual pair with prescribed complementarity nullity, and show that nullity correlates with solver difficulty. Our family applies the same idea to the NCM problem. Their generator does not apply directly: it builds a linear-objective SDP and chooses the constraint operator as part of the construction, whereas NCM fixes the constraint operator as $\text{diag}(X) = e$ and leaves only G free. As far as we have found, no NCM study uses instances with a known exact solution; the closest prior device is the pair of computable bounds on the distance $\|X^* - G\|_F$ of Higham and Strabić [21], which bracket the optimal *value* without exhibiting the optimal X^* , and so cannot serve as a forward-error reference. The test sets of [18, 31, 5, 6, 20, 2] are random or data-derived, as are the Anymatrix CORRINV matrices [19] used in [22], and in all of them the reference solutions are computed approximately.

Finite-precision line searches. An objective-value sufficient-decrease test cannot be evaluated accurately near a minimizer, because the difference of function values suffers cancellation. Hager and Zhang [15] introduced approximate Wolfe conditions for this reason. For the NCM dual, Borsdorf [5, §4.7.1] derived the stagnation threshold, $\|\nabla\theta\|$ of the order of the square root of the rounding error in θ (about 10^{-8}), noted that the problem is general to Armijo backtracking, and proposed the alternatives that became the modified step selection of [6]. Section 4 replicates this known phenomenon; it does not rediscover it.

Methods not compared here. A substantial augmented-Lagrangian and semismooth-Newton line of work addresses matrix-optimization problems adjacent to (1): the H -weighted nearest correlation matrix [30], the rank-constrained variant [13], and general convex conic and quadratic semidefinite programs [34, 24, 10]. Those methods are not included in the comparison of section 5. Specialized to the unweighted, rank-unconstrained (1), the H -weighted and augmented-Lagrangian formulations reduce to or embed the same semismooth Newton iteration this paper already tests; the general convex-conic and quadratic-SDP codes were not run here, and a matched-accuracy comparison against them on this instance family is left open, not ruled out. The instance family

itself is built from the KKT system of (1) alone and does not extend to those problems without a separate construction.

1.2 Outline

Section 2 constructs the instance family. Section 3 states the protocol, section 4 describes the two implementation traps, section 5 reports the ranking study, and section 6 examines the dimension-scaled stopping rule. Section 7 covers reproducibility and section 8 discusses limitations. The appendices contain the proofs for the instance family, the verification records for our reimplementations of AGD-SDAJ and Anderson-APM, the artifact manifest, and pseudocode for every solver.

2 The KKT-controlled instance family

2.1 Dual formulation

Following [27, 31], the Lagrangian dual of (1) is the unconstrained problem

$$\min_{y \in \mathbb{R}^n} \theta(y) := \frac{1}{2} \|\Pi_{\mathcal{S}_+^n}(G + \text{Diag}(y))\|_F^2 - e^\top y, \quad \nabla \theta(y) = \text{diag}(\Pi_{\mathcal{S}_+^n}(G + \text{Diag}(y))) - e. \quad (2)$$

Here $\Pi_{\mathcal{S}_+^n}$ is the projection onto the positive semidefinite cone. The gradient is globally Lipschitz with constant 1. If y^* satisfies $\nabla \theta(y^*) = 0$, then $X^* = \Pi_{\mathcal{S}_+^n}(G + \text{Diag}(y^*))$ is the unique solution of (1). Each evaluation of θ or $\nabla \theta$ requires one symmetric eigendecomposition (EVD) of $C(y) := G + \text{Diag}(y)$.

2.2 Construction and exactness

The idea is to choose the solution first and then build a G for which that solution satisfies the KKT conditions exactly. This is the standard technique of generating instances from prescribed optimality conditions [25, 9], and in particular Wei and Wolkowicz's generator of semidefinite programs with prescribed complementarity nullity [33], specialized here to the NCM problem. What is specific to NCM is the exact projection identity of proposition 2.2, the control of the rank, multiplicity and separation in section 2.3, and the entrywise conditions of section 2.4. Their construction does not cover the present problem: it generates the data of a *linear*-objective SDP and is free to choose the constraint operator itself (their Algorithm 2.3, Steps 3–4), whereas the NCM problem fixes both the quadratic objective and the operator $\mathcal{A} = \text{diag}$ with $b = e$, leaving G as the only free data. The shared element is the idea of prescribing an optimal primal–dual pair and controlling the dimension of its common nullspace, their complementarity nullity, which corresponds to $|\beta|$ here.

Definition 2.1 (KKT-controlled instance). Let $n > r \geq 1$. Draw $U \in \mathbb{R}^{n \times r}$ with independent standard-normal entries and rescale every row to unit Euclidean norm. Set $X^* = UU^\top$, so that $\text{diag}(X^*) = e$, $X^* \succeq 0$ and $\text{rank } X^* = r$ almost surely. Let $Q_0 \in \mathbb{R}^{n \times (n-r)}$ have orthonormal columns spanning $\ker X^*$. For $\mu \in \mathbb{R}_{\geq 0}^{n-r}$ define

$$W = Q_0 \text{Diag}(\mu) Q_0^\top, \quad G = X^* - W + \text{Diag}(\text{diag } W), \quad y^* = -\text{diag}(W). \quad (3)$$

Proposition 2.2 (Exactness). *For every $\mu \geq 0$, $\Pi_{\mathcal{S}_+^n}(G + \text{Diag}(y^*)) = X^*$ and $\nabla \theta(y^*) = 0$. Hence X^* is the unique nearest correlation matrix to G .*

Proof. By construction, $G + \text{Diag}(y^*) = X^* - W$. Both X^* and W are symmetric positive semidefinite, and $X^*W = X^*Q_0 \text{Diag}(\mu)Q_0^\top = 0$ because the columns of Q_0 lie in $\ker X^*$. The ranges of X^* and W are therefore orthogonal. They need not be complementary: if some $\mu_j = 0$, then $\text{range } W$ is a proper subspace of $\ker X^*$. Consequently $X^* - W$ has the spectral decomposition obtained by combining those of X^* and $-W$ on orthogonal subspaces. Its positive part is X^* , and its nonpositive part is $-W$. Thus $\Pi_{\mathcal{S}_+^n}(X^* - W) = X^*$, and $\nabla\theta(y^*) = \text{diag}(X^*) - e = 0$. Since θ is convex and differentiable, y^* minimizes (2). Uniqueness of the primal solution is classical [18, 31]. $\square$

In floating point, the generator recomputes the projection rather than relying on the identity. Over the 270 released instances at $n = 100$, the KKT residual is at most 1.7×10^{-14} and the exactness error $\|\Pi_{\mathcal{S}_+^n}(C(y^*)) - X^*\|_F$ at most 8.4×10^{-14} .

2.3 Prescribed spectral structure

Because the ranges are orthogonal, the spectrum of $C(y^*) = X^* - W$ is the multiset union

$$\text{spec } C(y^*) = \{\lambda_1(X^*), \dots, \lambda_r(X^*)\} \cup \{-\mu_1, \dots, -\mu_{n-r}\}. \quad (4)$$

In the notation of [31], let α, β, γ index the positive, zero and negative eigenvalues of $C(y^*)$. Then

- $|\alpha| = r$, so the rank of the solution is prescribed;
- $|\beta| = \#\{j : \mu_j = 0\}$ exactly, so the multiplicity of zero eigenvalues at the solution is prescribed;
- an entry $\mu_j = \delta > 0$ produces an eigenvalue $-\delta$, which moves toward the nonsmooth boundary of $\Pi_{\mathcal{S}_+^n}$ as $\delta \rightarrow 0$.

With the layout used below (m entries of μ equal to δ , the rest equal to $\mu_{\text{bulk}} > 0$), the multiplicity and the separation are therefore not varied independently in their effect on strict complementarity: $|\beta| = m$ when $\delta = 0$ and $|\beta| = 0$ otherwise.

Terminology. Qi and Sun [31] showed that constraint nondegeneracy always holds for the NCM problem, so the family does not, and could not, vary constraint degeneracy. What it varies is *failure of strict complementarity*: zero eigenvalues of $C(y^*)$ ($|\beta| > 0$), and near-zero eigenvalues (small δ) that approach it. Throughout this paper, a *degeneracy cell* means one (r, m, δ) combination in this sense, and the *strict-complementarity defect* means $|\beta|$. The generator uses $\mu = (\mu_{\text{bulk}}, \dots, \mu_{\text{bulk}}, \delta, \dots, \delta)$ with m copies of δ and $\mu_{\text{bulk}} = 0.1$. The separation parameter is $\text{sep} = \delta/\mu_{\text{bulk}}$ (we avoid ρ , which denotes the line-search backtracking factor in sections 4 and E). The released grid is

$$r \in \{5, 20, 50\}, \quad m \in \{1, 5, 20\}, \quad \delta \in \{10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}, 0\},$$

with five paired seeds per rank. This gives $3 \times 3 \times 6 = 54$ cells and 270 instances at $n = 100$. In plots, sep is shown on a $\log_{10}$ axis and the case $\text{sep} = 0$ is shown separately.

2.4 Correlation-like inputs: sufficient conditions for $|G_{ij}| \leq 1$

Proposition 2.2 places no restriction on the entries of G . For the instances to resemble invalid correlation matrices, which is the setting in which NCM problems arise, the generator additionally requires $\text{diag}(G) = e$ (true by construction), $\lambda_{\min}(G) < 0$, and $|G_{ij}| \leq 1$. The first two are checked

directly. The results below explain when the third can be expected to hold. Let P_r be the orthogonal projector onto $\text{range } U$, so that $Q_0 Q_0^\top = I - P_r$, and define

$$a := \max_{i \neq j} |X_{ij}^*|, \quad b := \max_{i \neq j} |(P_r)_{ij}|.$$

Proposition 2.3 (Deterministic bound, uniform μ). *If $\mu = c e$ with $c > 0$, then $|G_{ij}| \leq a + cb$ for all $i \neq j$. In particular, if $a < 1$, then $0 < c \leq (1 - a)/b$ is sufficient for $|G_{ij}| \leq 1$.*

Proposition 2.4 (Fixed-rank saturation). *Fix $r \geq 2$ and let the rows u_i of U be independent and uniform on S^{r-1} . Then $a \rightarrow 1$ as $n \rightarrow \infty$, along every realization.*

Proposition 2.4 says only that a saturates. It does *not* imply that a fixed $c > 0$ eventually violates proposition 2.3: at fixed r the quantity b is observed to decrease with n as well, but we do not prove this, so whether $a + cb$ exceeds 1 depends on the relative rates of $1 - a$ and b , which we do not establish. We therefore make no asymptotic feasibility claim, and the generator relies on entrywise screening.

Proposition 2.5 (High-dimensional regime). *Let the rows of U be independent and uniform on S^{r-1} and let $\eta \in (0, 1)$ be a failure probability (distinct from the near-zero value δ in μ). Then with probability at least $1 - \eta$,*

$$a \leq \sqrt{\frac{2}{r} \left(2 \log n + \log \frac{2}{\eta} \right)}.$$

In particular $a \rightarrow 0$ in probability whenever $r / \log n \rightarrow \infty$.

The bound follows from the explicit cap estimate of [3, Lem. 2.2] (subgaussianity in the sense of [32, Thm. 3.4.5] gives the same tail without a specific constant) and a union bound; see section A. We state it only for a , the quantity that governs proposition 2.3 in the regime of interest. A companion bound on b would require a model for $\text{range } U$ that we do not need here, since the released family lies in the fixed-rank regime of proposition 2.4, where proposition 2.5 does not apply.

Proofs are given in section A. These results give sufficient conditions only. Failure of the bound in proposition 2.3 does not show that some $|G_{ij}|$ exceeds 1, and for that reason the generator checks every entry directly instead of relying on the bound. Proposition 2.3 assumes uniform μ , whereas the released instances use a mixed μ (bulk value 0.1 together with m copies of δ). The extension to mixed μ is left open: the extra off-diagonal term it introduces depends on the choice of orthonormal basis for $\ker X^*$ and is not determined by (n, r, m, δ) alone (section A). The released mixed- μ instances are therefore screened entrywise rather than certified by a bound. None of these results says anything about indefiniteness of G , which is checked separately.

Which regime the released family is in. Table 1 reports empirical values of (a, b) at $n = 100$ over 30 seeds per rank. The family is in the regime of proposition 2.4, not proposition 2.5. The rank is fixed and small relative to n , a is close to 1 at $r = 5$, and the bound on c implied by proposition 2.3 shrinks sharply as the rank decreases.

The table is consistent with the screening outcomes. At $\mu_{\text{bulk}} = 0.1$, the sufficient bound at the mean values at $r = 5$ (0.162) exceeds 0.1, but at the worst seeds ($\max a = 0.9981$) the bound itself falls below 0.1. In the paired design (section 2.5), 5 of the 7 candidate seeds were accepted at $r = 5$; the 2 rejections were for $|G_{ij}| > 1$. At $r = 20$ and $r = 50$, 5 of 5 were accepted. With $\mu_{\text{bulk}} = 1$, the sufficient bound at the mean values still holds at $r = 20$ and $r = 50$ (1.690 and 2.650 exceed 1) but fails at $r = 5$, and in an earlier design every $r = 5$ candidate was rejected. Since one bulk level is used at every rank, this is why the generator uses $\mu_{\text{bulk}} = 0.1$.

Table 1: Empirical (a, b) at $n = 100$, 30 seeds per rank. The last column is the sufficient bound $(1 - a)/b$ of proposition 2.3 evaluated at the mean values of a and b .

r	mean a	max a	mean b	max b	$(1 - \bar{a})/\bar{b}$
5	0.9905	0.9981	0.0590	0.0708	0.162
20	0.7495	0.8340	0.1483	0.1871	1.690
50	0.5081	0.5719	0.1856	0.2083	2.650

2.5 Paired-seed design and screening

Each candidate seed is screened once per rank against its entire (m, δ) block. A seed is accepted only if all 3×6 instances built from it pass the checks ($\text{diag}(G) = e$, $|G_{ij}| \leq 1$, $\lambda_{\min}(G) < 0$, and a recomputed KKT residual and exactness error each below 10^{-9}). The accepted seeds are then reused across every (m, δ) cell. Varying the separation therefore changes only μ , with U held fixed, so comparisons along δ are within-seed. Rejections are counted and reported, not silently discarded.

Because of the screening, the $r = 5$ sample is conditioned differently from the $r = 20$ and $r = 50$ samples: at $r = 5$ it excludes the seeds whose instances have an off-diagonal entry above 1 (at $n = 100$, both rejections occurred at $m = 20$, $\delta = 10^{-2}$). These are not simply the seeds with the largest a , since the rejection depends on the basis-dependent kernel term as well. Comparisons across ranks should be read with this in mind.

Extent of the family. Every synthetic instance is close to feasible: $\lambda_{\min}(G)$ lies between -0.075 and -0.017 at $n = 100$ and between -0.047 and -0.011 at $n = 500$. A larger bulk level would give more strongly invalid inputs, but it fails the entrywise screen at $r = 5$, and the family uses a single bulk level at every rank; the degree of invalidity is therefore not a controlled factor here. Findings that depend on it (sections 5.8 and 5.9) rest on perturbed matrices whose reference is computed, not exact.

Regeneration. The seed of candidate k at rank r and dimension n is $100000n + 1000r + k$. The multiplicity is limited by the kernel dimension, $m \leq n - r$, and the m entries equal to δ are attached to the last m columns of Q_0 , the kernel block of the eigenvector basis of X^* returned by the symmetric eigensolver, which is the point at which a different LAPACK build can yield a different member of the family (section D).

2.6 Release

We release the 270 paired instances (G , X^* , y^* , SHA-256 hashes of the upper triangle of G , and per-instance screening diagnostics), the generator, and the seed-screening log. Section D lists the files. We also release a paired set at $n = 500$, generated on the same grid. All 270 instances passed screening; seed acceptance was 5 of 7 candidates at $r = 5$ (both rejections for $|G_{ij}| > 1$) and 5 of 5 at $r = 20$ and $r = 50$, with $\max_{i \neq j} |G_{ij}| = 0.99987$ among the accepted $r = 5$ instances. The ranking study of section 5.6 uses all five paired seeds per rank.

3 Protocol

This section sets out the measurement protocol. Each rule addresses a confound that we observed on this family, and each is stated together with the evidence that motivates it.

3.1 Native stopping rules as a confound

Suppose every solver is run to its own exit criterion and the resulting costs are compared. The comparison then credits a method for stopping early as much as for converging quickly. This happens even when all methods share one nominal rule. Under the dimension-scaled rule $\|\nabla\theta\|_2 \leq 10^{-7}n$ used by [22], which at $n = 100$ is 10^{-5} , the six solvers of section 5 and the Newton-SIN control stop at median forward errors between 5.04×10^{-8} and 1.93×10^{-5} , a factor of 383 (section 6, fig. 5). A cost comparison taken at these exits compares different accuracies.

3.2 Work–accuracy trajectories against the exact solution

We take one EVD as the unit of cost. After *every* EVD, whether it produces an iterate, a line-search trial or a Jacobian factorization, we record the pair

$$(\text{cumulative EVDs}, \|X_k - X^*\|_F),$$

together with $\|\nabla\theta\|_2$ and a flag indicating whether the evaluated point was accepted. Here X^* is the exact solution from proposition 2.2, not the output of another solver. All runs use a common, tight tolerance of 10^{-11} , and each solver has its own cap (section 5). These trajectories are work–precision diagrams in the sense of [16], and reading costs off them at a ladder of targets follows data profiles [29] and fixed-target evaluation [17]. The difference is that the reference is the exact solution, not the best value found by any solver. For any target ε , the cost can then be read off the trajectory after the run. We define the *cost to ε* as the cumulative EVD count at the first accepted iterate from which $\|X_k - X^*\|_F \leq \varepsilon$ holds for every later accepted iterate, that is, at the start of the final sub- ε stretch of the run. A solver whose last accepted iterate lies above the target is recorded as not reaching it at that ε . A run that dips below the target, leaves it, and settles below it again is credited from the point at which it settles.

Trajectories also expose *plateaus*. The cost of a quadratically convergent method takes very few distinct values across many decades of ε : in table 5, Newton-SIN-BH costs 3, 4 or 5 EVDs at every target from 10^{-2} to 10^{-10} . A single-tolerance comparison therefore depends on where the tolerance falls relative to these plateaus.

3.3 Accepted-only accounting

A rejected line-search trial always costs one EVD. Under *accepted-only* accounting it can never be credited with reaching a target, because the method did not move to that point. Under *all-trial* accounting it can, which credits a method for an accuracy it never returned. We use accepted-only accounting throughout and report the other convention as a sensitivity check (section 5.11); on the family studied here the difference turns out to be small.

3.4 Primitive work as the unit of cost

Table 2 gives the median primitive work at matched tolerance (10^{-11}) over the 270 instances.

Four accepted Newton iterations use 5 EVDs and 24 Krylov iterations. SBB-Dual, Dykstra-APM and Anderson-APM use one EVD per iteration and no further primitive, although Anderson-APM’s

Table 2: Median primitive work at matched tolerance 10^{-11} , 270 instances, $n = 100$. An ‘outer iteration’ denotes a different amount of work in each row.

solver	EVDs	CG iterations	line-search trials	accepted outer its
Newton-SIN-BH	5	24	0	4
Newton-SIN	11	29.5	0	5
AGD-SDAJ-BH	70	0	14	9
AGD-SDAJ	81.5	0	25	10
SBB-Dual	46.5	0	0	45.5
Dykstra-APM	102.5	0	0	102.5
Anderson-APM	34	0	0	34

least-squares update carries $O(mn^2)$ work that the EVD count does not record. Nine AGD-SDAJ-BH outer iterations use 70 EVDs, because each outer iteration contains $q = 2$ quasi-Newton inner steps and one accelerated gradient step, each with its own line search, and the accelerated step can need a further evaluation (algorithm 2). A table of ‘iterations to convergence’ across these rows would compare four different quantities. We therefore report EVDs, Krylov iterations and line-search trials separately.

3.5 Timing protocol

All timings use Julia with `BLAS.set_num_threads(1)`. A warm-up solve is run and discarded, `GC.gc()` is called before the measured solve, and a single `@elapsed` surrounds the whole solve. A separate instrumented pass (TimerOutputs) gives the breakdown of time. In that breakdown, percentages are taken against the measured elapsed time, so any uninstrumented work shows up as ‘other’ time instead of inflating the EVD share. Timings were measured on all 270 instances at $n = 100$, and on an 18-instance subset at $n = 500$ (listed in `code/n500_timing_subset.txt`), at both the native ($10^{-7}n$) and the matched (10^{-11}) tolerance. The trajectory runs are a separate, untimed pass.

All seven variants were timed back to back in a single session, because timings from different sessions are not comparable: an earlier session on the same machine, with identical EVD counts on every run, measured the dual methods, whose code had not changed, between 3.7 and 4.0 times slower per EVD at $n = 100$. Each method is timed in an implementation that does not add work of its own. The projection methods rebuild the projection from the positive eigenpairs, as the dual methods do, rather than from all n , which removes an $O(n^3)$ product per EVD and gives bit-identical iterates on all 1104 runs checked; and Anderson-APM is timed in a form that updates its buffers in place, with iterates bit-identical to the transcription verified in section C. Timed as transcribed, it would have cost 2.3 to 3.8 times more per EVD. All timings were measured on one machine: Intel Core 7 150U (10 physical cores, 12 logical), 32 GB RAM, Windows 11 (build 26200), Julia 1.12.6 with OpenBLAS (ILP64) through libblastrampoline. The timing runs pin `BLAS.set_num_threads(1)`; the untimed trajectory pass at $n = 500$ was run with four BLAS threads, which changes wall-clock only and not the EVD counts it records.

3.6 Fairness of the evd as a unit of work

The EVD count is a good proxy for cost only if one EVD costs about the same in every method. Table 3 and fig. 1 give the median wall-clock time per EVD at matched tolerance.

The median is taken over every run, including any that exhausted its budget without converging

Table 3: Microseconds per EVD at matched tolerance (10^{-11}), $n = 100$, single-threaded BLAS.

solver	$\mu\text{s}/\text{EVD}$	relative to cheapest
SBB-Dual	1133	1.00
AGD-SDAJ-BH	1181	1.04
AGD-SDAJ	1188	1.05
Dykstra-APM	1278	1.13
Anderson-APM	1323	1.17
Newton-SIN	1341	1.18
Newton-SIN-BH	1695	1.50

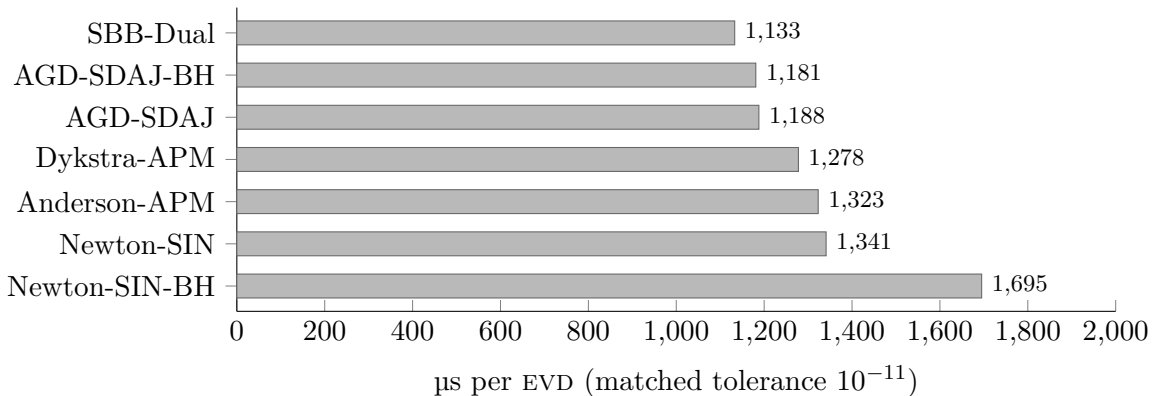


Figure 1: Median wall-clock time per EVD at $n = 100$, all seven variants timed in one session. The spread is a factor of 1.50; Newton’s EVDs are the dearest, because they are accompanied by Krylov solves that the EVD count does not record. Anderson-APM’s EVDs cost 1.17 times SBB-Dual’s, the difference coming from its work outside the eigendecomposition, chiefly its least-squares update of $O(mn^2)$ work per EVD.

(Newton-SIN’s 56 naive-Armijo stalls of section 4.2 foremost among them); in these timing tables their cost to that exit counts as their cost, unlike the cost-to-target rule of section 3.2, under which a run that never reaches a target is recorded as not reaching it. Table 3 shows a factor-of-1.50 spread in per-EVD cost across the seven variants, and the largest per-EVD cost is Newton’s: each Newton outer iteration includes a Krylov solve that no EVD count records, so counting EVDs understates Newton’s cost and flatters it. Two examples bound how much this matters here. First, Newton-SIN-BH is cheaper than SBB-Dual by a factor of 9.3 in EVDs (5 against 46.5) but only by a factor of 5.4 in wall-clock time (0.009 s against 0.050 s): the EVD count overstates Newton-SIN-BH’s advantage, but does not reverse it. Second, Anderson-APM spends about 16% of its time outside the eigendecomposition (its least-squares update alone about 13% at $n = 100$ and 15% at $n = 500$), against about 1% for SBB-Dual, so its EVDs cost 1.17 times SBB-Dual’s. The bias runs the same way as Newton’s: Anderson-APM needs 1.37 times fewer EVDs than SBB-Dual but only about 1.05 times less time, so the EVD count again overstates the advantage of the method with fewer EVDs without reversing it. In this single session, then, the known bias in EVD-counting is smaller than the gaps it is used to compare. We report both metrics throughout; whether this single-session picture survives repeated measurement is the question the next two paragraphs answer.

Session-to-session variation. Table 3 is a single measured session. Repeating the matched-tolerance timing measurement four further times, on the same instances and the same code, shows that a given solver’s median microseconds/EVD varies far more across sessions than the solvers differ from each other within one session: by a factor of between 3.8 and 4.9, depending on the solver, at $n = 100$, and between 4.5 and 4.9 at $n = 500$, across these five sessions (`results/analysis/timing_repeats.md`; five sessions need not have sampled the extremes, so the true spread may be larger), against the $\leq 1.50\times$ spread between solvers found above. Two features of the ranking nonetheless survive this noise. The cheapest end is consistent: the cheapest solver per EVD is always SBB-Dual or an AGD variant, at both dimensions. The dearest-per-EVD end is consistent too, though not unanimous: Newton-SIN-BH holds it at $n = 100$ in four of the five sessions (Anderson-APM in the fifth) and at $n = 500$ in four of five (Dijkstra-APM in the fifth, by about 1.6% over Anderson-APM). It is the middle of the ranking that reorders between sessions, which is why the specific ratios in table 3 — 1.04, 1.05, 1.13, 1.17, 1.18 — should be read as one session’s draw rather than as stable constants, and not as an ordering among the solvers they describe. This session-to-session noise does not touch the 9.3-in-EVDs comparison above, since that comparison involves no timing measurement at all. It affects the 5.4-in-time comparison only mildly: a slowdown that hits every solver cancels in a ratio, and across the five sessions at $n = 100$ the per-EVD ratio of Newton-SIN-BH to SBB-Dual ranged from 1.36 to 1.50, so that figure should be read to about ten per cent.

The cost metric agrees in half the sessions, and always picks the same winner. At matched tolerance, ranking the seven timed variants by median EVDs and by median wall-clock time gives the same order in this session:

$$\begin{aligned} \text{Newton-SIN-BH} &< \text{Newton-SIN} < \text{Anderson-APM} < \text{SBB-Dual} \\ &< \text{AGD-SDAJ-BH} < \text{AGD-SDAJ} < \text{Dijkstra-APM}. \end{aligned}$$

Repeating the comparison across the four further sessions above, the two rankings agree exactly in 2 of the 5 sessions at $n = 100$ and 3 of 5 at $n = 500$ (`results/analysis/timing_repeats.md`), 5 of 10 in all; since EVD counts are identical across sessions by construction, every disagreement comes from the wall-clock side. Two things are nonetheless robust across all ten sessions (five at each dimension). First, Newton-SIN-BH is cheapest overall under either cost metric, every time. Second, the trailing position is usually but not always robust: Dijkstra-APM is dearest overall in 8 of the 10 sessions — all five at $n = 500$, but only three of five at $n = 100$, where in two sessions AGD-SDAJ overtakes it as the most expensive by wall-clock time despite needing about a fifth fewer EVDs (81.5 against 102.5). So only the leading position is unconditionally stable; the trailing one is merely usual, and the middle reorders. Where the ranking does move, it is not scattered: apart from that trailing swap, the solver overtaken is always Anderson-APM, by SBB-Dual at $n = 100$ in every session where the two metrics disagree, and at $n = 500$ by AGD-SDAJ-BH (and, in one session, AGD-SDAJ). Anderson-APM’s per-EVD premium alone does not predict which sessions flip, so this is a further source of instability in the ranking, alongside the accuracy target and the instance regime. What the ten sessions do establish is that the identity of the single cheapest method does not depend on the choice of cost metric. Even within one session the metric can decide a close pair: at $n = 500$, Anderson-APM needs fewer EVDs than AGD-SDAJ-BH on 13 of 18 instances but less time on only 11.

3.7 Summary of the protocol

1. Measure forward error against an exact X^* .

2. Run every solver to a common tight tolerance and record a trajectory point after every EVD.
3. Read costs off the trajectories at a common target; give rejected trials no accuracy credit.
4. Report EVDs, Krylov iterations, line-search trials and wall-clock time, never iterations alone.
5. Report the feasibility of the returned iterate ($\lambda_{\min}(X)$ and $\|\text{diag}(X) - e\|_{\infty}$), and state which iterate is returned.
6. Report reach rates beside every median that is computed over a subset of instances.

4 Two implementation traps

This section describes two implementation defects. Neither prevents convergence to the correct solution at the tolerances usually reported, so neither is visible from final errors. Both change measured cost substantially. We present them as evidence for the protocol of section 3. They are not new discoveries.

4.1 Trap 1: redundant Jacobian factorization

A semismooth Newton step at y_k needs the spectral decomposition of $C(y_k)$ to apply the generalized Jacobian [31]. That decomposition was already computed when $\theta(y_k)$ and $\nabla\theta(y_k)$ were evaluated, at the accepted trial of the previous line search. An implementation that recomputes it pays one extra EVD per outer iteration and gains nothing. On the literature matrix `cor1399` (section 5.10), where both variants take the same six Newton steps, recomputing the decomposition raises Newton’s cost from 7 to 13 EVDs, an increase of about 86% (a factor of 1.86), with an identical final error (`results/timing_session1/timing_real_matched.csv`). Newton-SIN-BH caches the decomposition throughout this paper; Newton-SIN, the naive-Armijo control of section 4.2, does not, so its EVD counts already include this redundant factorization on top of the naive test’s own cost. The same class of defect turned up in our first reimplement of AGD-SDAJ (section B, defect 1). It was found in the same way, by counting primitives one at a time against a published count.

4.2 Trap 2: the finite-precision Armijo test

An objective-value sufficient-decrease test,

$$\theta(y_k + \alpha d_k) \leq \theta(y_k) + c\alpha \nabla\theta(y_k)^{\top} d_k, \quad (5)$$

becomes a plain comparison of two rounded copies of θ_k once the predicted decrease $c\alpha |\nabla\theta(y_k)^{\top} d_k|$ falls below half an ulp of θ_k : the right-hand side then rounds to θ_k itself, and the test degenerates to $\text{fl}(\theta_t) \leq \text{fl}(\theta_k)$. It stops judging the step correctly once the achieved decrease is itself buried in that same rounding error, near θ_k . Near the solution $|\nabla\theta^{\top} d| \gtrsim \|\nabla\theta\|^2$ (equality for a gradient-type step; for Newton’s step it can be considerably larger), so this second, more severe threshold falls at a gradient norm of roughly $\sqrt{u|\theta|}$, where u is the unit roundoff. Past that point, whether a trial passes is decided by rounding noise rather than by the step’s real merit: a step that improves both the gradient and the forward error can still be rejected, and the search must halve until it happens to land where the noise lets a trial through. In this regime Newton does not halt outright; it creeps, accepting a string of steps far smaller than warranted until it exhausts its iteration budget without converging — a stall in the sense used below, as the counts just below show. None of

Table 4: AGD-SDAJ on `degen-r20-m1-d1e-10-p2`, as published and with the Borsdorf–Higham adaptation.

variant	outer	EVDs	exit	final $\ \nabla\theta\ $	$\ X - X^*\ _F$
AGD-SDAJ (as published)	115	4001	EVD budget	3.60×10^{-9}	6.09×10^{-9}
AGD-SDAJ-BH	10	67	converged	9.40×10^{-12}	1.82×10^{-11}

this is new. The cancellation problem is well known in general optimization [15]. For the NCM dual, Borsdorf [5, §4.7.1] derived the threshold (about 10^{-8} in $\|\nabla\theta\|$ when $|\theta| \approx 1$) and proposed alternatives, which became the modified step selection of Borsdorf and Higham [6, Sec. 3.3, Alg. 4.1]. Their Section 3.3 observes that when $|\phi(x)| = 1$, $\|p\|_2 < u^{1/2}$ and $\|\nabla\phi(x)\|_2 < u^{1/2}/2$, the computed values $fl(\phi(x+p))$ and $fl(\phi(x))$ coincide, so that the Armijo condition cannot be verified at all. Their Algorithm 4.1, Step 6, then tests whether the two objective values are *nearly equal*, using $|a - b| < \gamma u(1 + |a| + |b|)$ with $\gamma = 100$, and if so accepts the full inexact Newton step provided $\|\nabla f(y_k + d_k)\|_2 / \|\nabla f(y_k)\|_2 \leq 1 - \mu$, falling back to a unit steepest-descent step otherwise. Our implementation uses their parameter values, $\mu_{\text{BH}} = 0.9$, $\rho = 0.5$ and $\sigma = 10^{-4}$ (we write μ_{BH} for their gradient-reduction parameter, since μ denotes the spectrum vector of Definition 2.1).

Semismooth Newton. With the naive test (5), Newton stalls (exits at `max_iter` without converging) on 56 of the 270 KKT instances; across all 270 instances it uses 151,471 backtracks in total. With Borsdorf–Higham globalization (parameters $\mu_{\text{BH}} = 0.9$, $\sigma = 10^{-4}$, halving, and $\gamma = 100$ in the near-equality test), it converges on all 270 with 8 backtracks. The count of 56 was reproduced exactly in a separate timing run of the same code: at matched tolerance, the naive-Armijo control exits at `max_iter` on 56/270. Both counts are in the shipped result files: the 56 is the `exit` column of the `Newton-SIN` rows of `timing_kkt270_matched.csv`, and the 8 is the `linesearch_trials` total of the `Newton-SIN-BH` rows of that file, which agrees with the `backtracks` column of `ranking_kkt270_v2.csv.gz`. The backtrack total of 151,471 is the same run’s own count of rejected trials, cross-checked against its recorded per-trial trajectory (`code/trace_armijo_stall.jl`). Of the 5,677 rejected trials of the undamped Newton step ($\alpha = 1$, before any backtracking), every one, without exception, would, if accepted, have reduced both the gradient norm and the exact forward error; counting every rejected trial, including those reached only after backtracking, 115,211 of the 151,471 (76%) would have.

AGD-SDAJ. The same failure occurs in AGD-SDAJ [22], a separately published method with a different algorithmic structure. Line 5 of Algorithm 1 in that paper is the objective-value test (5). Run to the common tolerance 10^{-11} , AGD-SDAJ as published exhausts the 4000-EVD budget on 18 of the 270 instances (the budget is checked at the start of each outer iteration, so a stalled run ends at between 4000 and 4035 EVDs). On instance `degen-r20-m1-d1e-10-p2`, the trajectory contains 3198 rejected monotone line-search trials, with the forward error $\|X_k - X^*\|_F$ and $\|\nabla\theta\|$ fixed at 6.0935×10^{-9} and 3.5986×10^{-9} . The predicted decrease is about 1.3×10^{-21} , far below the rounding error in θ , which is about $u|\theta| \approx 3 \times 10^{-14}$ because $\theta(y^*) \approx 3 \times 10^2$ on this instance. We adapted the Borsdorf–Higham near-equality test to Algorithm 1’s step: when the objective difference cannot be resolved, a trial is accepted if it reduces $\|\nabla\theta\|$. With this change, all 18 stalls disappear (table 4). Across the family, the number of instances reaching the tolerance rises from 252/270 to 270/270, and the worst-case cost falls from 4035 EVDs to 131.

4.3 Scope of these findings

Three points limit what these results show.

1. *The phenomenon is known.* Borsdorf analysed it for the NCM dual [5, §4.7.1], and the fix was published in [6]. What we add is limited to (i) its recurrence in a separately published method that uses a naive Armijo test, where it is invisible under that method’s native stopping rule, and (ii) its quantification on the controlled family (56/270 and 18/270 stalls). For semismooth Newton the remedy is the published Borsdorf–Higham modified line search; for AGD-SDAJ it is our adaptation of their near-equality safeguard to the Algorithm 1 step of that method.
2. *It does not affect Huynh and Hwang’s reported results.* Their stopping rule, $\|\nabla\theta\|_2 \leq 10^{-7}n$, equals 10^{-5} on these instances and stops the method far above the precision floor at which the test fails. The stall appears only when the method is pushed to a tight common tolerance, which is what a matched-accuracy comparison requires. On their own instances, our reimplementations reproduces their published counts exactly for P8 and P9, and matches P7 in outer iterations and residual with one differing backtracking decision (section 7, section B).
3. *The adaptation is ours.* AGD-SDAJ-BH is our modification and is not specified by the original authors. We therefore report it alongside the unmodified method on every test set where both were run (all except the 135 higher-rank instances of section 5.7).

The general point is that whether a method appears to converge at a given tolerance can depend on a line-search detail that is unrelated to the algorithm’s design. A benchmark that stops each method under its own rule never exposes this.

5 Ranking under different conventions

5.1 Solvers

We compare six variants of five methods. All are implemented in Julia with a common evaluation kernel, and every evaluation of θ or $\nabla\theta$ costs exactly one EVD. Unless stated otherwise, each variant starts from $y_0 = 0$ (Dykstra-APM starts from $X_0 = G$). The algorithms below are written at the same level of detail and use the notation $(\theta_k, g_k, X_k) \leftarrow \text{EVAL}(y_k)$ for one EVD of $C(y_k)$, with $X_k = \Pi_{\mathcal{S}_+^n}(C(y_k))$ and $g_k = \text{diag}(X_k) - e$. The stopping criterion in all algorithms is $\|g_k\|_2 \leq \tau$ (Dykstra-APM and Anderson-APM, which have no dual iterate, apply it to the diagonal of their PSD iterate). Here $\tau = 10^{-11}$ unless stated otherwise.

Newton-SIN-BH (algorithm 1) is the semismooth Newton method of Qi and Sun [31]. The Newton system is solved inexactly by conjugate gradients, using the block form of the generalized Jacobian restricted to the positive-eigenvalue columns (with $\Omega_{\beta\beta} = 0$) and a 10^{-12} diagonal shift. The spectral decomposition from the accepted trial is cached (section 4.1), and globalization uses the Borsdorf–Higham modified line search [6, Alg. 4.1, Step 6]. We adopt their globalization only: their Algorithm 4.1 also solves the Newton system with preconditioned MINRES rather than CG, which we do not adopt, so Newton-SIN-BH is not a reimplementations of their algorithm. *Newton-SIN* is the same method with the naive Armijo test in place of the Borsdorf–Higham globalization, and without the cached factorization of section 4.1 (so its EVD counts carry both traps at once); it appears only as a control in sections 3 and 4.

AGD-SDAJ (algorithm 2) is Algorithm 4 of Huynh and Hwang [22], transcribed from the full text and checked against the published counts (section B). It uses $q = 2$ inner quasi-Newton steps with a diagonal secant preconditioner and the nonmonotone residual line search of their equation (6), followed by one accelerated gradient step with a monotone Armijo line search. *AGD-SDAJ-BH* is our adaptation of that Armijo step, described in section 4.2.

SBB-Dual (algorithm 3) is a spectral (Barzilai–Borwein) gradient method on (2). It uses the BB1 step, clamped to $[\eta, 2 - \eta]$ with $\eta = 0.01$, has no line search, performs one EVD per update, and finishes with a diagonal-rescaling step $\hat{X} = D^{-1/2} X D^{-1/2}$, $D = \text{Diag}(\text{diag } X)$, as in [6].

Dykstra-APM (algorithm 4) is Higham’s alternating projections with Dykstra’s correction [18]. Each iteration uses one EVD. The method returns the PSD half-iterate.

Anderson-APM (algorithm 5) is Dykstra-APM with Anderson acceleration, as given by Higham and Strabić [20]. The fixed-point map is one alternating-projections step on the pair $(Y, \Delta S)$, and the acceleration solves a least-squares problem over the last m differences of its residuals, taken here at the authors’ default $m = 2$. It costs one EVD per iteration plus $O(mn^2)$ work, and like Dykstra-APM it is scored on the PSD half-iterate. Our implementation transcribes their released code. Its iterates agree with an independent implementation to 2.3×10^{-12} , and it reproduces their published iteration counts where rounding at their stopping tolerance does not decide them. Under `syevr`, the eigensolver used for every result in this paper, it matches 57 of 64 small-matrix counts; switching only the eigensolver to `syevd` matches both large-matrix counts exactly and 61 of 64 small ones, with the rest one iteration apart, confirming the remaining gap under `syevr` is rounding at the stopping tolerance rather than a transcription error; on one ill-conditioned matrix the counts are decided by rounding and differ by up to 56 iterations (section C).

5.2 Design

All six variants were run on the 270 instances of section 2 at $n = 100$, to a common tolerance of 10^{-11} , and a trajectory was recorded as described in section 3.2. Each solver ran under its own cap: 4000 EVDs for the Newton and AGD variants, 20000 iterations for SBB-Dual and 2000 for the two projection methods. Only unmodified AGD-SDAJ reached its cap at $n = 100$ (section 4.2). We evaluated 11 conventions: the native exit; a common forward-error target $\|X - X^*\|_F \leq \varepsilon$ for $\varepsilon \in \{10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}\}$; and a common dual residual $\|\nabla \theta\|_2 \leq \tau$ for the same five values. We also compared accepted-only with all-trial accounting. Section 5.6 repeats the study at $n = 500$.

5.3 Cost of the semismooth Newton method across conventions

The 11 conventions produce three distinct orderings, and Newton-SIN-BH is cheapest in all three: it leads under every accuracy target, under both error measures, under either accounting rule and at every solver’s own exit, and it is strictly cheapest on every one of the 270 instances at 10^{-8} and 10^{-10} . In median, Anderson-APM is second under 10 of the 11 conventions and tied for second, with both AGD variants, under the loosest dual-residual target. Per instance its hold is much weaker: it is strictly cheapest after Newton-SIN-BH on 136 of the 270 instances at 10^{-8} and 139 at 10^{-10} , or on 157 and 142 counting ties. The conventions change cost ratios and the order of the remaining places; they never change which method is cheapest. The rest of this section is about those remaining places, which is where the choice of test set decides the answer.

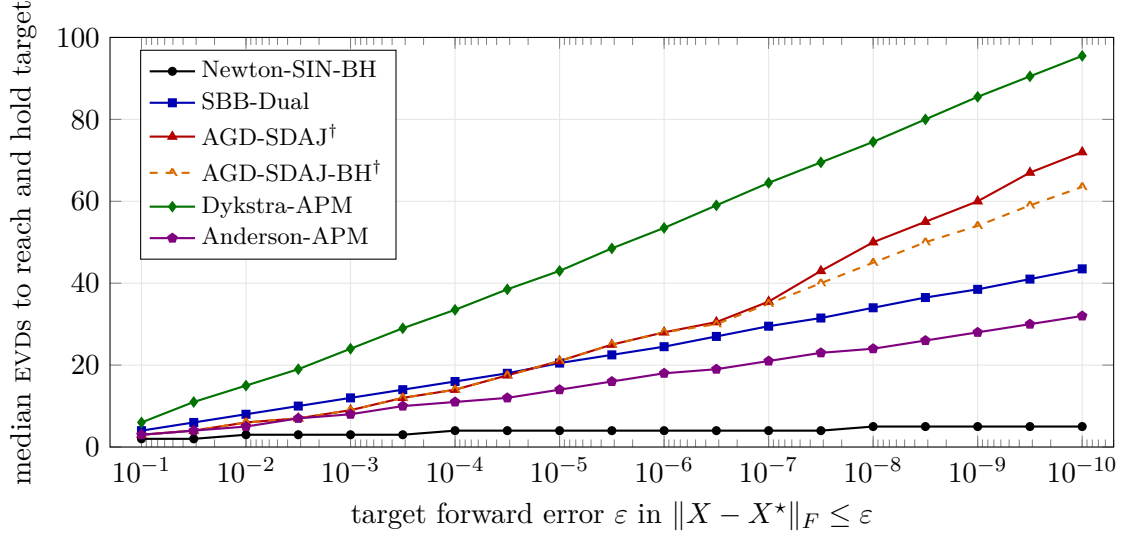


Figure 2: Median cumulative EVDs to reach and hold $\|X_k - X^*\|_F \leq \varepsilon$ (accepted-only accounting) over 270 instances, $n = 100$, on a half-decade grid. Among the plotted solvers, Anderson-APM is second at every target from 10^{-2} on, and tied with both AGD variants at the two loosest. SBB-Dual and AGD-SDAJ-BH cross between $\varepsilon \approx 3 \times 10^{-5}$ and 10^{-5} . [†]AGD-SDAJ-BH and every other solver reach and hold each target shown on all 270 instances. Unmodified AGD-SDAJ does so on all 270 down to 3×10^{-7} , and on 253/270 at 10^{-10} ; below 3×10^{-7} its medians are taken over the instances it reached. The curves stop at 10^{-10} . Cluster-bootstrap intervals for the decade targets are given in table 5; they are narrower than the plotted marks at most targets.

5.4 Dependence of the ordering on the accuracy target

Table 5 and fig. 2 show median cost to a common target. Anderson-APM is second at every target, from 5 EVDs at 10^{-2} to 32 at 10^{-10} , where AGD-SDAJ-BH needs 63.5 and SBB-Dual 43.5. The places below it move. In the pooled medians, both AGD variants are cheaper than SBB-Dual at the two loosest targets (10^{-2} , 10^{-4}), and SBB-Dual is cheaper from 10^{-6} onward. The same reversal occurs at the same point under the forward-error and the dual-residual criteria, so it does not depend on which error measure is used. A paper that reports at a loose tolerance would place AGD-SDAJ above SBB-Dual, and a paper that reports at a tight tolerance would place them the other way. Both are defensible one-number summaries of the same runs, although, as the paired comparison below shows, the two methods are level at 10^{-4} and the change of order is concentrated in one rank.

Reach and hold. A target counts as reached at the first accepted iterate from which the error stays at or below the target for the rest of the run. The accelerated methods are not monotone in forward error, so this differs from the first iterate at which the error dips below the target; scoring the first dip alone would understate their cost. Under this rule Newton-SIN-BH, Anderson-APM, AGD-SDAJ-BH, SBB-Dual and Dykstra-APM reach every target on all 270 instances. Only unmodified AGD-SDAJ misses some: 264/270 and 253/270 at forward targets 10^{-8} and 10^{-10} (264 and 254 for the residual), which is consistent with its 18 budget-exhausting stalls (section 4.2). Its entries marked * are medians over the instances it reached, and are therefore optimistic. The third-place reversal, between SBB-Dual and the AGD variants, is the same whether or not those

Table 5: Median EVDs to reach and hold $\|X - X^*\|_F \leq \varepsilon$ (accepted-only) at $n = 100$, with cluster-bootstrap 95% intervals in brackets ($B = 2000$, resampling the five paired seeds within each rank, so that all 18 (m, δ) cells of a drawn seed move together: 15 clusters in all). The medians pool the three ranks with equal weight, and the $r = 5$ seeds are screened differently from the others (section 2.5). Rows are ordered by the 10^{-8} column. Entries marked * are medians over the subset of instances that reached the target, the subset size in small type; all other entries are over 270/270.

solver	10^{-2}	10^{-4}	10^{-6}	10^{-8}	10^{-10}
Newton-SIN-BH	3 [3, 3]	4 [3, 4]	4 [4, 4]	5 [5, 5]	5 [5, 5]
Anderson-APM	5 [5, 5]	11 [11, 11]	18 [17, 18.5]	24 [24, 26]	32 [30, 34]
SBB-Dual	8 [8, 8]	16 [15.5, 16]	24.5 [24.5, 24.5]	34 [34, 34]	43.5 [43, 43.5]
AGD-SDAJ-BH	6 [6, 6.5]	14 [13, 15]	28 [27, 29]	45 [42, 48.5]	63.5 [62, 66]
AGD-SDAJ	6 [6, 6.5]	14 [13, 15]	28 [27, 29]	50* [46, 52] [264]	72* [70, 75] [253]
Dykstra-APM	15 [15, 15]	33.5 [33.5, 33.5]	53.5 [53.5, 53.5]	74.5 [74.5, 74.5]	95.5 [95.5, 95.5]

entries are included. A first-hit rule (the first accepted iterate with error at or below the target, whether or not it holds) leaves four of the six solvers’ medians unchanged at every target, and moves Anderson-APM on 5 instances at 10^{-2} (by 2 EVDs each, median unaffected). It lowers the two AGD variants’ medians at the two tightest targets: AGD-SDAJ-BH from 45 to 43 at 10^{-8} and from 63.5 to 62.5 at 10^{-10} ; AGD-SDAJ from 50 to 48 (also reaching 266 instead of 264) and from 72 to 70 (257 instead of 253). The disagreement is on 53 to 62 of 270 instances at those two targets (one instance each at 10^{-6} , not enough to move the median), 5 to 8 EVDs apart there in median. Both shifts leave AGD-SDAJ-BH and AGD-SDAJ well above SBB-Dual’s 34 and 43.5, so table 5’s median ordering does not depend on requiring sustained accuracy rather than a first crossing, though the exact AGD medians do.

Uncertainty. The bootstrap intervals in table 5 resample 15 clusters, so they describe the variation between seeds and are narrow by construction; they do not test the ordering. What does is the paired per-instance comparison, which is much closer than the pooled medians suggest and which we report here as the median paired difference in EVDs (SBB-Dual minus AGD-SDAJ-BH), its cluster interval, and the share of the 270 instances on which SBB-Dual is cheaper: +2 [1, 2], 40.4% at 10^{-2} ; +1.5 [−1, 2], 48.5% at 10^{-4} ; −4 [−5, −4], 54.1% at 10^{-6} ; −11 [−13.5, −9], 61.9% at 10^{-8} ; and −19 [−22, −17], 65.6% at 10^{-10} . Two objections could be raised against reading these numbers as evidence of an ordering, and both need addressing before we say what the numbers do show. First, an instance-level sign test would treat the 270 comparisons as independent although they in fact form only 15 clusters, so it would overstate the evidence at every target; we therefore do not report one. Second, the cluster interval quoted above is a better guide but is not free-standing evidence either: the bootstrap resamples the five seed clusters within each rank separately and always keeps the three ranks in equal proportion, so its sign is fixed whenever every cluster within a rank already agrees on direction, as happens here. Both objections trace to the same underlying fact, which we can state directly. Within a rank the sign of the paired median does not change with the target at $r = 5$ (AGD-SDAJ-BH cheaper, by 13.5 to 97.5 EVDs) or at $r = 50$ (SBB-Dual cheaper, by 1 to 26); it changes only at $r = 20$, between 10^{-4} (+1.5) and 10^{-6} (−5). From 10^{-6} on, then, the pooled interval excludes zero because two of the three ranks agree with each other and outnumber the third under equal weighting, not because 15 exchangeable clusters were resampled and happened to concur: resampling all 15 together without stratifying by rank spans zero at 10^{-6} , 10^{-8} and 10^{-10} alike. The finding that survives is therefore the per-rank one just stated, not the

Table 6: Orderings by cell at forward target 10^{-8} (54 cells, five paired instances each), with the number of cells of each rank producing each ordering.

ordering (cheapest first)	cells	$r=5$	$r=20$	$r=50$
Newton-SIN-BH < Anderson-APM < AGD-SDAJ-BH < AGD-SDAJ < SBB-Dual < Dykstra-APM	16	15	1	0
Newton-SIN-BH < SBB-Dual < Anderson-APM < Dykstra-APM < AGD-SDAJ-BH < AGD-SDAJ	13	0	2	11
Newton-SIN-BH < Anderson-APM < SBB-Dual < AGD-SDAJ-BH < AGD-SDAJ < Dykstra-APM	12	0	12	0
Newton-SIN-BH < SBB-Dual < Anderson-APM < AGD-SDAJ-BH < AGD-SDAJ < Dykstra-APM	7	0	1	6
Newton-SIN-BH < Anderson-APM < AGD-SDAJ-BH < SBB-Dual < AGD-SDAJ < Dykstra-APM	1	0	1	0
Newton-SIN-BH < SBB-Dual < Anderson-APM < AGD-SDAJ-BH < Dykstra-APM < AGD-SDAJ	1	0	1	0
Newton-SIN-BH < AGD-SDAJ < AGD-SDAJ-BH < Anderson-APM < SBB-Dual < Dykstra-APM	1	1	0	0
Newton-SIN-BH < AGD-SDAJ-BH < AGD-SDAJ < Anderson-APM < SBB-Dual < Dykstra-APM	1	1	0	0
Newton-SIN-BH < AGD-SDAJ-BH < Anderson-APM < AGD-SDAJ < SBB-Dual < Dykstra-APM	1	1	0	0
Newton-SIN-BH < SBB-Dual < Anderson-APM < Dykstra-APM < AGD-SDAJ < AGD-SDAJ-BH	1	0	0	1

pooled interval read as an independent confirmation of it. Read this way, the pooled numbers above say: the two methods are level at 10^{-4} (win share 48.5%, essentially even); AGD-SDAJ-BH is cheaper at 10^{-2} by a modest median margin with substantial per-instance overlap; and SBB-Dual is cheaper in the pooled median from 10^{-6} on, but only on 54.1%, 61.9% and 65.6% of the instances at 10^{-6} , 10^{-8} and 10^{-10} respectively. The dependence of the pooled comparison on the target is thus entirely the rank dependence of section 5.5, weighted equally across the three ranks by the grid; its crossing point between 10^{-4} and 10^{-6} is a property of that grid, not of the methods alone. At $n = 500$ (section 5.6), SBB-Dual is cheaper than AGD-SDAJ-BH on 0 of 270 instances at 10^{-2} .

5.5 Dependence of the ordering on the degeneracy parameters

Aggregate medians can hide dependence on the regime. At a fixed target (forward error 10^{-8}), the 54 (r, m, δ) cells produce 10 distinct orderings (table 6). This description is exploratory: each cell holds five instances, no correction for multiple comparisons is applied to the cell-level orderings, and where two cell medians are equal the order follows the fixed listing of the solvers. That affects seven cells in which SBB-Dual and Anderson-APM tie exactly (six at $r = 50$, one at $r = 20$), so the count of 10 depends on the tie-break. What is claimed below is the rank effect, which is present in every one of the five seeds.

Newton-SIN-BH is cheapest in every cell. Below it, at $n = 100$, the order is governed almost entirely by the rank r of the solution. AGD-SDAJ-BH is cheaper than SBB-Dual in all 18 cells with $r = 5$; SBB-Dual is cheaper in 16 of the 18 cells with $r = 20$ and in all 18 with $r = 50$. Within a rank, the near-zero multiplicity m and the separation δ rarely change this pair's order. Both AGD variants fall below Dykstra-APM to last place in 14 cells: all 12 cells with $r = 50$ and $m \in \{5, 20\}$, and the two cells with $r = 20$, $m = 20$ and $\delta \in \{10^{-8}, 10^{-10}\}$. This block is determined by rank and

multiplicity; the separation plays no visible role in it. The instance family was built to vary these quantities one at a time. On a collection of real matrices with uncontrolled spectra, the ordering could not be associated with rank in this way.

A crossover between a published method and our adaptation of one. The reversal is not confined to SBB-Dual. Dykstra-APM, a published method, and AGD-SDAJ-BH, our Borsdorf–Higham adaptation of a published method (section 4.2), also exchange places with rank, between $r = 20$ and $r = 50$ rather than between $r = 5$ and $r = 20$. At $r = 5$ AGD-SDAJ-BH is cheaper on all 90 instances at 10^{-8} (median difference +223.5 EVDs) and at $r = 20$ on 80 of 90, with one tie; at $r = 50$ Dykstra-APM is cheaper on 61 of 90 (median -3.5, cluster-bootstrap 95% interval $[-6, -2]$) and on 75 of 90 at 10^{-10} ($-8.5, [-12, -5]$); with only 5 seed clusters at this rank, these within-rank intervals are close to the range of the 5 seed medians, so the evidence for the direction is really the unanimous sign across those 5, not an independent interval on top of it. All five seeds agree on the sign at $r = 5$ and at $r = 50$, though 5-of-5 agreement alone gives a two-sided sign-test $p = 0.0625$, above the conventional 0.05 threshold for either rank taken alone; it is the same sign recurring at $r = 50$ and, oppositely, at $r = 5$, on two independently seeded sets of instances at different ranks, that gives the pooled 10-of-10 pattern its strength ($p \approx 0.002$). At $r = 50$ alone the evidence does not clear the conventional threshold, and the margin is small, about an eighth of AGD-SDAJ-BH’s median cost of 28 EVDs: 26 of the 90 instances go the other way and 3 tie at 10^{-8} . The reversal is therefore reproducible without SBB-Dual, on a pair in which only AGD-SDAJ-BH is the author’s adaptation, not a new method of the author’s own. It does not occur by the same rank at higher dimension: at $n = 500$ AGD-SDAJ-BH is still ahead at $r = 150$ (Dykstra-APM cheaper on 13 of 45 at 10^{-10} , section 5.7). On the equity panel at $n = 2105$ the corresponding pair is Anderson-APM and AGD-SDAJ-BH, whose order is not monotone in rank: Anderson-APM is cheaper at $r = 50$ and $r = 400$ and AGD-SDAJ-BH at $r = 100$ and $r = 200$ (section 5.9).

The location of the SBB-Dual/AGD-SDAJ-BH crossover is nevertheless specific to $n = 100$. At $n = 500$ (sections 5.6 and 5.7) SBB-Dual is behind at every rank up to $r = 50$ and ahead from $r = 75$: both the direction of the effect and the crossover survive, but the crossover moves to a higher rank.

That rankings depend on the required accuracy is expected in principle [29, 17]. What the controlled family adds is that the variation is organized by a single spectral quantity which the construction sets directly.

Each cell holds five instances, so the count of distinct orderings overstates how much genuinely varies. Resampling the paired seeds of each cell ($B = 2000$) leaves the adjacent pair AGD-SDAJ-BH against AGD-SDAJ unresolved in most cells, as expected of two methods differing only in a safeguard that rarely activates at $n = 100$; reading unresolved pairs as ties collapses much of table 6. What survives is the placement of SBB-Dual relative to the AGD pair, resolved in the great majority of cells. Figure 3 shows the same comparison cell by cell as a ratio of median costs, which separates the three controlled quantities: rank sets the level, the near-zero multiplicity modulates it within a rank, and the separation has almost no effect.

5.6 The same design at $n = 500$

The instance family was regenerated at $n = 500$ with the same grid. All 270 instances screened successfully, with the same acceptance pattern as at $n = 100$: 5 of 7 candidate seeds at $r = 5$ and 5 of 5 at $r = 20$ and $r = 50$. The ranking study uses all five paired seeds of each rank, 270 instances, with caps of 1200 iterations for SBB-Dual and 800 for Dykstra-APM. It was run in two parts, seeds 0–2 (about 8.6 h) and seeds 3–4 (about 7.1 h), with identical code and settings.

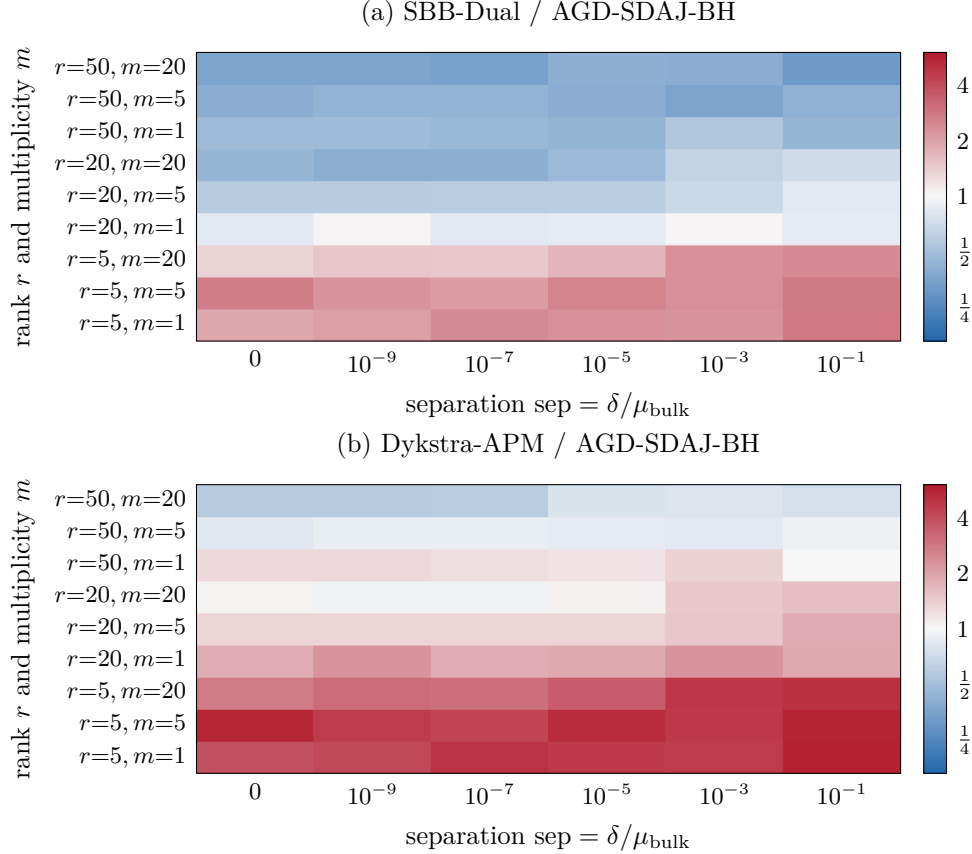


Figure 3: Ratio of median EVDS to reach and hold $\|X - X^*\|_F \leq 10^{-8}$ for each of the 54 cells at $n = 100$ (five paired instances per cell), on one shared logarithmic colour scale: red cells favour AGD-SDAJ-BH, blue cells the other method, white is parity. (a) SBB-Dual over AGD-SDAJ-BH. The ratio is organized primarily by rank: it exceeds 1 in every $r = 5$ cell and is below 1 in every $r = 50$ cell. The near-zero multiplicity m modulates the level within a rank, most visibly at $r = 20$ (from 0.84–1.03 at $m = 1$ to 0.40–0.71 at $m = 20$), whereas the separation has little effect. (b) Dykstra-APM over AGD-SDAJ-BH, a published method and our adaptation of another. The organization is the same, because Dykstra-APM costs between 1.6 and 2.7 times SBB-Dual in every cell, so the whole map is shifted towards AGD-SDAJ-BH: the ratio lies between 2.7 and 5.7 at $r = 5$, between 0.94 and 2.2 at $r = 20$ and between 0.58 and 1.35 at $r = 50$, and it falls below 1 in 14 cells, all 12 with $r = 50$ and $m \in \{5, 20\}$ and two with $r = 20$ and $m = 20$.

On this grid ($r \leq 50$), two of the $n = 100$ conclusions survive and two do not; section 5.7 shows that the two that fail return at higher rank.

Confirmed. Newton-SIN-BH is again cheapest at every target, at 3 to 5 EVDS. The stall of unmodified AGD-SDAJ recurs (31 of 270 runs exhaust the EVD budget, against none for AGD-SDAJ-BH, whose worst case is 240 EVDS against 4039 for unmodified AGD-SDAJ; the budget is checked between outer iterations, not within a line search, so an exhausting run can overshoot the round 4000-EVD cap by up to one line search’s trials before it halts).

Not reproduced for $r \leq 50$. The change of order between SBB-Dual and the AGD variants seen in table 5 is absent: at $n = 500$ AGD-SDAJ-BH is ahead of SBB-Dual at all five targets. Only Anderson-APM moves, from fourth at 10^{-2} to second from 10^{-4} on (table 7). The paired

Table 7: $n = 500$, 270 instances. Median EVDs to reach and hold $\|X - X^*\|_F \leq \varepsilon$, with cluster-bootstrap 95% intervals ($B = 2000$, resampling paired seeds within rank) and reach counts where these are below 270. The Dykstra-APM row is censored (see text): its medians cover 270 instances at 10^{-2} and 10^{-4} and 180 from 10^{-6} , which is why 195 at 10^{-4} exceeds 191 at 10^{-6} .

solver	10^{-2}	10^{-4}	10^{-6}	10^{-8}	10^{-10}
Newton-SIN-BH	3 [3, 3]	4 [3, 4]	4 [4, 4]	5 [5, 5]	5 [5, 5]
Anderson-APM	13 [12, 14.5]	23.5 [23, 28]	40.5 [36.5, 43.5]	53.5 [53, 55]	70.5 [67.5, 74]
AGD-SDAJ-BH	9 [9, 9]	30 [29, 32]	53.5 [50.5, 56]	79 [76, 80]	101.5 [98.5, 105]
AGD-SDAJ	9 [9, 9]	30 [29, 32]	55.5 [52, 58] ^[268]	87 [84, 89] ^[259]	119 [117, 123] ^[242]
SBB-Dual	44 [44, 44]	97 [97, 97]	152 [152, 152]	207.5 [207, 208]	260 [259, 262] ^[261]
Dykstra-APM [‡]	87 [86.5, 88]	195 [194.5, 195.5]	191 [191, 192] ^[180]	258 [258, 260] ^[180]	325.5 [325, 327.5] ^[180]

per-instance comparison is one-sided. SBB-Dual is cheaper on 0 of 270 instances at 10^{-2} , 3 at 10^{-4} and 15 at 10^{-8} . We report these counts and the cluster-bootstrap intervals of table 7 rather than an instance-level significance test: the design reuses each drawn U across all 18 (m, δ) cells of its rank, so the 270 comparisons form 15 seed clusters, and a test treating them as independent would overstate the evidence by a wide margin. The rank effect of section 5.5 keeps its direction but no longer produces a crossover: the median paired difference at 10^{-8} falls from +669.5 EVDs at $r = 5$ to +129 at $r = 20$ and +13 at $r = 50$, where SBB-Dual is cheaper on 17% of instances. Cell by cell, SBB-Dual is ahead of AGD-SDAJ-BH in 0 of the 54 cells (47 with a bootstrap interval resolved in favour of AGD-SDAJ-BH, 7 unresolved); with 5 seed clusters per cell, a resolved interval here is again close to unanimous agreement across those 5, not independent confirmation on top of it. SBB-Dual is thus behind at every rank tested at $n = 500$, while it led at $r \geq 20$ at $n = 100$. The shrinking gap, however, points to a crossover beyond the grid, which section 5.7 confirms.

This is not an artefact of the caps. SBB-Dual converged on all 270 instances (worst case 1075 EVDs against its 1200 cap) and reached every target except 10^{-10} on nine instances. [‡]Dykstra-APM, by contrast, *is* censored: 90 of 270 runs stop at the 800-iteration cap, and from 10^{-6} down its medians cover only 180 instances. Its position at $n = 500$ should be read as censored rather than measured.

The scaling is the cleaner way to state the difference. Between $n = 100$ and $n = 500$ the median cost to 10^{-8} rises by a factor of 6.1 for SBB-Dual (34 to 207.5 EVDs) and by 1.8 for AGD-SDAJ-BH (45 to 79), while Newton-SIN-BH is unchanged at 5. The order below first place therefore reflects the test set, including its dimension, rather than a fixed property of the methods.

5.7 Higher ranks at $n = 500$

The grid of section 2 stops at $r = 50$. To locate the crossover we generated 135 further instances at $n = 500$ with $r \in \{75, 100, 150\}$, $m \in \{1, 5, 20\}$, $\delta \in \{10^{-4}, 10^{-8}, 0\}$, $\mu_{\text{bulk}} = 0.1$ and five paired seeds per rank, all accepted by screening, and ran every variant except unmodified AGD-SDAJ (table 8).

Between $r = 50$, where SBB-Dual is behind by 13 EVDs at 10^{-8} (section 5.6), and $r = 75$, where it is ahead by 10, the order reverses. The dependence on the accuracy target returns with it: at $r = 75$ and $r = 100$ the two methods are level at 10^{-4} (median differences +1 and 0), with AGD-SDAJ-BH ahead for $m = 1$, while SBB-Dual is ahead at 10^{-8} and 10^{-10} . Both patterns of section 5.5 therefore survive at $n = 500$; what changes with dimension is the rank at which they appear, from between $r = 5$ and $r = 20$ at $n = 100$ to between $r = 50$ and $r = 75$ at $n = 500$ (fig. 4). We report these as brackets and fit no trend to them: the synthetic family covers only two

Table 8: $n = 500$, higher ranks, 45 instances per rank. Median EVDs to reach and hold $\|X - X^*\|_F \leq 10^{-8}$, and the number of instances (out of 45) at 10^{-8} on which SBB-Dual, and on which Anderson-APM, is cheaper than AGD-SDAJ-BH. All runs converged.

r	median EVDs to 10^{-8}					cheaper at 10^{-8} , of 45	
	Newton	Anderson	SBB-Dual	AGD-SDAJ-BH	Dykstra	SBB vs AGD	Anderson vs AGD
75	5	26	51	60	110	34	45
100	4	22	37	52	81	43	45
150	5	18	22	41	52	45	45

dimensions, each with three or four rank points, which cannot separate proportional growth from any other increasing rule, and the equity instances of sections 5.8 and 5.9 sit outside the synthetic family in more ways than dimension. Newton-SIN-BH is strictly cheapest on all 135 instances at every target from 10^{-4} to 10^{-10} .

A pilot of 24 instances at $r \in \{100, 150\}$, each repeated with $\mu_{\text{bulk}} = 0.01$ on the same X^* (48 runs in all) and hence a tenfold smaller violation, gave the same result: at 10^{-8} the median paired difference was -12.5 against -14 EVDs at $r = 100$ and -19.5 against -18 at $r = 150$. The two medians are close (within 1.5 EVDs of each other at both ranks), and 48 instances at two ranks is not powered to detect a small residual effect of μ_{bulk} ; the pilot rules out a large one and is consistent with the effect following the rank of the solution rather than the size of W , not a confirmation that the two are identical.

5.8 Instances built from equity returns

Two further test sets start from one real correlation matrix: the sample correlation matrix C of daily log returns, computed from split-adjusted closing prices, of 550 US-traded equities over 1397 trading days (January 2020 to July 2025), with no missing values. The panel holds 50 names under each of the eleven sector labels of its source files and includes some ADRs and one closed-end fund. C is a valid correlation matrix, with $\lambda_{\min}(C) = 0.019$. It serves here only as a realistic input, not as an estimate of market structure; one series (CHRD) contains a corporate-action splice, which does not affect that use.

KKT instances seeded by the data. Because C has full rank it cannot itself be the solution of a nontrivial instance of section 2, whose construction needs a kernel. Its rank- r factor model can: with $C \approx V_r \Lambda_r V_r^T$, set $U = V_r \Lambda_r^{1/2}$, scale its rows to unit norm and take $X^* = UU^T$. The construction of section 2 then gives G with X^* as its *exact* nearest correlation matrix, whose eigenvectors come from the market data rather than from a random U . Replicates differ by a Haar-random rotation of the kernel basis. Two settings differ from the synthetic family, both forced by screening: at $r = 5$ the factor model misses 95% of the off-diagonal correlation and drives some $|X_{ij}^*|$ to 0.99995, so every $r = 5$ instance fails $|G_{ij}| \leq 1$, and with $\mu_{\text{bulk}} = 0.1$ every rank fails it. We therefore use $r \in \{20, 50, 100\}$ (off-diagonal relative fit 0.59, 0.39, 0.24) and $\mu_{\text{bulk}} = 0.01$, with $\delta \in \{10^{-5}, 10^{-9}, 0\}$ so that $\text{sep} = \delta/\mu_{\text{bulk}}$ stays on the synthetic grid, $m \in \{1, 5, 20\}$ and three rotations: 81 instances. All pass screening, with worst exactness 1.1×10^{-12} . They are only mildly invalid, $\lambda_{\min}(G) \in [-4.6, -1.2] \times 10^{-3}$.

Anderson-APM is again second at every rank (108, 64 and 53 EVDs to 10^{-8}), and it is cheaper than AGD-SDAJ-BH on 15, 25 and 23 of the 27 instances at $r = 20, 50$ and 100, on matrices whose eigenvectors come from market data rather than from a random orthogonal basis.

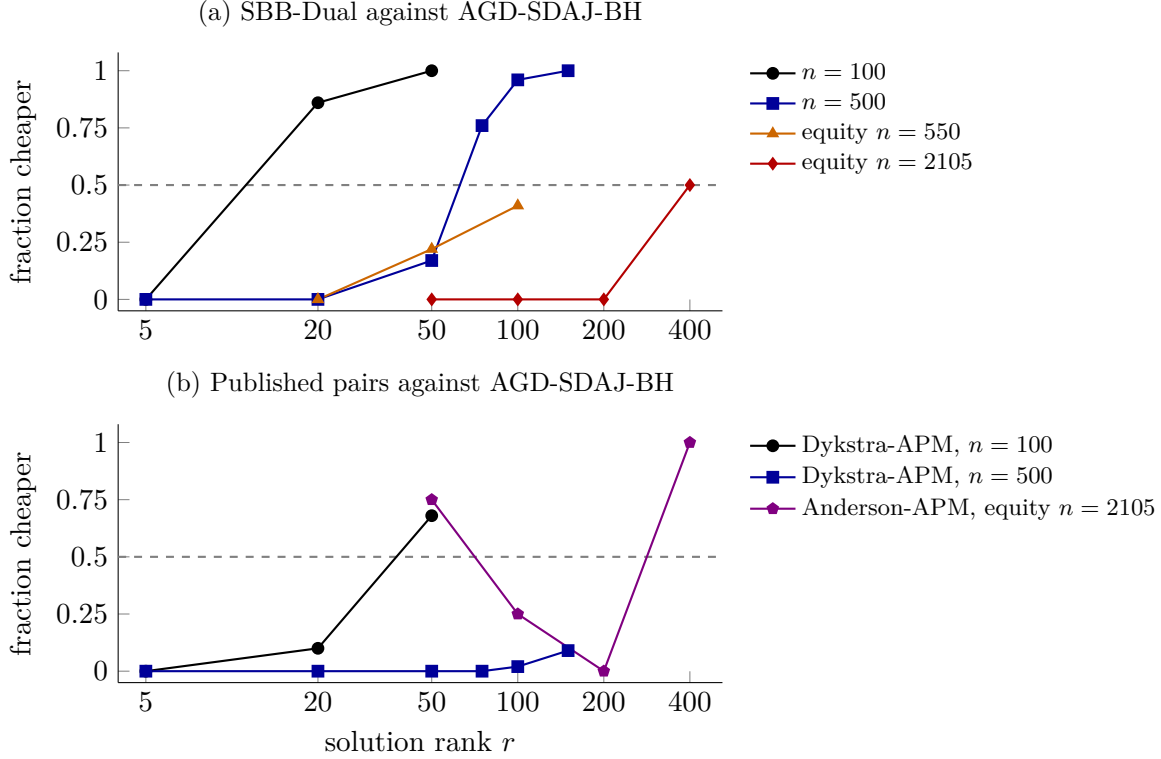


Figure 4: Where pairs of methods exchange places. Each point is the fraction of instances at one rank on which the named method reaches $\|X - X^*\|_F \leq 10^{-8}$ in fewer EVDS than AGD-SDAJ-BH; a run that never reaches the target counts as the more expensive one, and the dashed line marks parity. (a) SBB-Dual: the exchange happens between $r = 5$ and $r = 20$ at $n = 100$, between $r = 50$ and $r = 75$ at $n = 500$, and has not been reached by $r = 100$ on the equity instances at $n = 550$ or by $r = 200$ at $n = 2105$; the rightmost point rests on four instances and its ordering reverses at 10^{-10} (section 5.9). (b) Without SBB-Dual: Dykstra-APM overtakes AGD-SDAJ-BH between $r = 20$ and $r = 50$ at $n = 100$ but not by $r = 150$ at $n = 500$. At $n = 2105$, where each point rests on four instances, Anderson-APM is ahead at $r = 50$, behind at $r = 100$ and $r = 200$, and ahead again at $r = 400$, so its order is not monotone in rank. For SBB-Dual the rank at which the exchange occurs rises with dimension, and for Dykstra-APM the exchange has not occurred by $r = 150$ at $n = 500$; the SBB-Dual exchange is reported as a bracket per dimension and no trend is fitted to it.

Table 9 repeats the pattern of the synthetic family on data-derived spectra. Newton-SIN-BH is cheapest on all 81 instances at every target. Unmodified AGD-SDAJ exhausts its 4000-EVD budget on 10 of 81, and AGD-SDAJ-BH on none. Rank again organizes the comparison between SBB-Dual and AGD-SDAJ-BH: the median paired difference at 10^{-8} falls from +128 EVDS at $r = 20$ to +17 at $r = 50$ and +7 at $r = 100$. At $r = 100$ the gap nearly closes: SBB-Dual is cheaper on 11 of 27 instances at 10^{-8} and 13 of 27 at 10^{-10} ; at 10^{-8} it is ahead on all nine with $m = 1$ and behind on all nine with $m = 20$. On synthetic instances of similar size SBB-Dual is already ahead at $r = 75$ (section 5.7), so eigenvectors taken from market data appear to move the crossover to a still higher rank. The replicates share X^* within a rank, so these are counts, not independent samples.

Table 9: KKT instances seeded by equity returns, $n = 550$, 27 instances per rank. Median EVDs to reach and hold $\|X - X^*\|_F \leq 10^{-8}$ against the exact X^* ; bold marks the cheapest, and a bracketed count is the number of instances that reached the target where fewer than 27 did. Out of 27 instances, SBB-Dual is cheaper than AGD-SDAJ-BH on 0, 6 and 11 at $r = 20, 50$ and 100 (0, 6 and 13 at 10^{-10}), and Anderson-APM is cheaper than AGD-SDAJ-BH on 15, 25 and 23 (15, 25 and 24 at 10^{-10}).

r	Newton-SIN-BH	Anderson-APM	AGD-SDAJ-BH	AGD-SDAJ	SBB-Dual	Dykstra-APM
20	4	108	117	154	249	500
50	4	64	92	94.5 [24]	112	228
100	4	53	65	79 [25]	72	148

Table 10: Perturbed equity correlation matrix, $n = 550$, five draws per σ . Median EVDs to reach and hold forward error 10^{-8} against the computed reference, and the number of draws (out of 5) on which SBB-Dual is cheaper than AGD-SDAJ-BH.

σ	Newton-SIN-BH	SBB-Dual	Anderson-APM	Dykstra-APM	AGD-SDAJ-BH	AGD-SDAJ	SBB cheaper
0.005	5	10	11	17	20	24	5
0.01	5	15	18	30	20	30	5
0.03	6	29	31	60	29	29	2 (1 tie)
0.1	7	67	59	140	38	42.5 [4]	0

Perturbed data matrix. The second set follows test problem P12 of [22]: $G = \text{clip}(C + \sigma E, -1, 1)$ with unit diagonal, where E is symmetric with standard normal off-diagonal entries, for $\sigma \in \{0.005, 0.01, 0.03, 0.1\}$ and five draws each. The smallest eigenvalue of G falls from about -0.09 to -4.2 and the number of negative eigenvalues rises from about 60 to 240; at $\sigma = 0.1$ about 0.02% of entries are clipped. No exact solution exists, so forward error is measured against a Newton reference with $\|\nabla\theta\|_2 \leq 1.2 \times 10^{-13}$ on every matrix.

Here the degree of invalidity reverses the order below first place (table 10); the most negative eigenvalue and the number of negative eigenvalues grow together with σ , so the two cannot be separated. For small perturbations SBB-Dual is second, 1.3 to 2 times cheaper than AGD-SDAJ-BH in median, and at $\sigma = 0.005$ even Dykstra-APM is ahead of both AGD variants; at $\sigma = 0.03$ the two are level; at $\sigma = 0.1$ AGD-SDAJ-BH is cheaper on every draw, by a median factor of about 1.7. The same reversal holds at 10^{-4} and 10^{-6} . It is consistent with the matrices of section 5.10, where the mildly invalid Rocky Mountain matrix favours SBB-Dual and the larger, more strongly invalid matrices favour AGD-SDAJ-BH. Newton-SIN-BH is cheapest on every matrix at every target from 10^{-4} to 10^{-8} , and unmodified AGD-SDAJ stalls once, at $\sigma = 0.1$.

5.9 An equity panel at $n = 2105$

The panel of section 5.8 is one of 550 names over 1397 days. To test the same questions an order of magnitude higher we built a second panel of 2105 US-traded common stocks over 1253 trading days (September 2021 to September 2026), selected from an exchange listing by complete price history, a median traded value above \$1 M per day and one share class per company. Here n exceeds the sample length, so the sample correlation matrix C is singular: 853 of its eigenvalues are zero to working precision, and C sits exactly on the boundary of the feasible set.

Two sets were generated from it. The perturbed set adds σE as in section 5.8 for $\sigma \in \{0.005, 0.01, 0.03, 0.1\}$ with two draws each; the resulting matrices are strongly invalid, with $\lambda_{\min}(G)$

Table 11: Equity panel at $n = 2105$, four instances per rank. Median EVDs to reach and hold $\|X - X^*\|_F \leq 10^{-8}$ against the exact X^* , and the number of instances on which SBB-Dual is cheaper than AGD-SDAJ-BH at 10^{-8} and at 10^{-10} . A dash marks a method that did not reach the target within its cap of 800 EVDs.

r	Newton-SIN-BH	Anderson-APM	AGD-SDAJ-BH	AGD-SDAJ	SBB-Dual	Dykstra-APM	SBB cheaper
50	5	147.5	159.5	201.5	650	–	0, 0
100	5	233	143	179	442	–	0, 0
200	5	158.5	111	116.5	244	491	0, 0
400	5	78	109.5	130	109	222.5	2, 1

from -0.35 to -8.7 and 599 to 970 negative eigenvalues. The per-instance values behind these ranges are shipped (section D). The KKT set uses the rank- r factor model of C as X^* with $r \in \{50, 100, 200, 400\}$, $m \in \{1, 20\}$, $\delta = 0$ and two kernel rotations, 16 instances with exact solutions. At this size $\mu_{\text{bulk}} = 0.1$ passes the screen at every rank, so these instances use the same bulk level as the synthetic family.

Newton-SIN-BH takes 5 EVDs on every one of these 16 instances, and its median at the 10^{-8} target is 4 or 5 at $n = 100$ and $n = 500$ too (individual instances there range from 4 to 11): its median EVD count is flat in n over a twenty-fold range, while the first-order methods grow with n at very different rates, from about 2.4-fold for AGD-SDAJ-BH to about 10-fold for SBB-Dual; wall-clock time is not flat for either, since each EVD itself costs more at larger n (section 3.6). The gap between SBB-Dual and AGD-SDAJ-BH again narrows with rank, from $+490.5$ EVDs at $r = 50$ to -0.5 at $r = 400$ at the 10^{-8} target. We do not claim a crossover at $r = 400$: with only four instances at this rank (two rotations, two multiplicities, weak replicates rather than independent seed clusters), no cluster-bootstrap interval applies, so we report the raw range instead. The median there is -0.5 EVDs across a range of $[-37, +27]$, and at 10^{-10} the sign turns over ($+8$ across a range of $[-17, +26]$, with AGD-SDAJ-BH cheaper on three of the four instances). The crossover is therefore not reached by $r = 200$ and is unresolved at $r = 400$.

The two rotations per cell are weak replicates for several methods. Newton-SIN-BH’s costs are identical across rotations in every cell and SBB-Dual’s in six of eight; only the AGD variants vary with the rotation throughout. At $r = 200$ and $r = 400$ the costs of Anderson-APM and Dykstra-APM also differ between rotations by at most two EVDs, although Anderson-APM’s do vary at $r = 50$ and $r = 100$.

Anderson-APM converges on all 16 instances, and with AGD-SDAJ-BH it forms the one pair here, neither of them SBB-Dual, whose order changes with rank. It is cheapest after Newton-SIN-BH at $r = 50$ and $r = 400$, but AGD-SDAJ-BH is cheaper at $r = 100$ and $r = 200$. At 10^{-8} and 10^{-10} AGD-SDAJ-BH is cheaper on 4 of 4 instances at $r = 200$, by 33 to 70 EVDs, and Anderson-APM on 4 of 4 at $r = 400$, by 6 to 68; because Anderson-APM is nearly rotation-invariant at these ranks, each count rests on two distinct instances, one per multiplicity. Near-zero multiplicity also matters here in a way it did not at $n = 100$: at $r = 100$ the two instances with $m = 20$ favour AGD-SDAJ-BH by 176 to 211 EVDs, while for $m = 1$ Anderson-APM is ahead by 27 to 33 on one instance and within 3 EVDs of AGD-SDAJ-BH on the other (2 behind at 10^{-8} , 3 ahead at 10^{-10}). With four instances per rank these are descriptive counts, not estimates.

Dykstra-APM was run on the same 16 instances at a cap of 800 EVDs, one EVD costing about 1.8s at this size. It is last at every rank and target it reaches, so it changes no ordering, but its own behaviour across the set is the sharpest rank dependence we observed. At $r = 50$ and $r = 100$ it exhausts the cap, ending at forward errors of 2.5×10^{-6} to 8.9×10^{-6} and 2.9×10^{-8} to 6.7×10^{-8} ; at $r = 200$ and $r = 400$ it converges on its own diagonal-feasibility test, in 647 to 662 and 301 to

Table 12: Perturbed equity panel at $n = 2105$, two draws per σ . EVDs to reach and hold forward error 10^{-8} against a computed reference with $\|\nabla\theta\|_2 \leq 3.2 \times 10^{-13}$.

σ	Newton-SIN-BH	SBB-Dual	AGD-SDAJ-BH	AGD-SDAJ
0.005	6, 6	19, 19	34, 22	34, 76
0.01	6, 6	29, 28	32, 29	33, 34
0.03	7, 7	71, 67	37, 44	38, 48
0.1	8, 8	165, 162	68, 66	47, 72

305 EVDs. Its cost to a loose target falls monotonically with rank, from 556 EVDs at $r = 50$ to 93 at $r = 400$ for $\varepsilon = 10^{-4}$, a sixfold drop within one test set, while Newton-SIN-BH stays at 4. A study drawing its instances only from low-rank solutions would report Dykstra-APM as a method that does not converge at this size; one drawing from high-rank solutions would report it reaching 10^{-10} in 289 EVDs.

The perturbed set reproduces the pattern of table 10 at four times the dimension (table 12): SBB-Dual is second at the two smallest σ and about twice as expensive as AGD-SDAJ-BH at $\sigma = 0.1$, with the change of order between $\sigma = 0.01$ and $\sigma = 0.03$ in both panels. Unmodified AGD-SDAJ exhausted its 800-EVD budget on two of the 16 KKT instances, both at $r = 400$, where AGD-SDAJ-BH converged in 119 and 140 EVDs; this is the stall of section 4.2 at the largest size we tested.

Where Anderson-APM is not second. Its second place is a property of the KKT families, not of the method. On the perturbed equity matrices it is third at the two smallest σ , level with both AGD variants in a three-way tie at $\sigma = 0.03$, and behind both at $\sigma = 0.1$ (fourth of the five, per table 10); it is cheaper than AGD-SDAJ-BH on all five draws at $\sigma \in \{0.005, 0.01\}$ and on none at $\sigma \in \{0.03, 0.1\}$, so the degree of invalidity reorders it exactly as it reorders SBB-Dual. On the four literature matrices it is second on the Rocky Mountain matrix (6 EVDs), but fourth on both `cor1399` (103 against 66 for AGD-SDAJ-BH) and `cor3120` (155 against 41), the two matrices where every projection method is slow (table 13). A comparison drawn only from the exact families would therefore overstate its standing, consistent with the regime dependence reported throughout this section.

5.10 Four matrices from the literature

The family of section 2 is synthetic. To check that the regime dependence above is not an artefact of the construction, we ran the same protocol on four invalid correlation matrices from the literature: the Rocky Mountain carbon-storage matrix ($n = 94$), `cor1399` ($n = 1399$), `cor3120` ($n = 3120$) and `bccd16` ($n = 3250$), all from the Anymatrix `CORRINV` collection [19] used by [22].

No exact solution is available here, which is the situation the rest of this paper avoids. Forward error is measured against a reference computed by semismooth Newton driven towards 10^{-13} ; for `cor1399` that reference solve itself stopped at its iteration limit after 1770 EVDs with $\|\nabla\theta\|_2 = 4.59 \times 10^{-13}$. At $n \approx 3000$ an absolute tolerance of 10^{-13} is not attainable in double precision, so for `cor3120` and `bccd16` the reference solve is capped at 30 EVDs and its achieved residual is reported instead: $\|\nabla\theta\|_2 = 1.25 \times 10^{-12}$ and 5.03×10^{-13} , after 2506s and 444s in the recorded check run (`results/validation/reference_check.log`). A second reference for each, differently initialized (SBB-Dual driven to its floor, then the same Newton polish that produced the first reference) rather than by an independent algorithm, agrees with the first to $\|X_A - X_B\|_F = 1.3 \times 10^{-11}$ and

Table 13: Invalid correlation matrices from the Anymatrix **CORRINV** collection [19], run to a common tolerance of 10^{-9} . EVDs to reach and hold forward error 10^{-8} , total EVDs at the solver’s own exit, and the forward error there, all measured against a computed reference solution, not an exact one. [‡]Reaches its cap and is censored. Five solvers come from one run; Anderson-APM was run later against a recomputed reference, and a replay of AGD-SDAJ-BH against that reference reproduces every cost to target in this table.

matrix	solver	EVDs to 10^{-8}	total EVDS	final error
Rocky Mountain $n = 94$	Newton-SIN-BH	4	4	1.49×10^{-13}
	Anderson-APM	6	7	2.52×10^{-10}
	SBB-Dual	7	8	1.53×10^{-10}
	Dykstra-APM	11	13	2.57×10^{-10}
	AGD-SDAJ-BH	12	17	3.47×10^{-10}
	AGD-SDAJ	12	25	3.47×10^{-10}
cor1399 $n = 1399$	Newton-SIN-BH	7	7	1.27×10^{-9}
	AGD-SDAJ-BH	66	69	1.98×10^{-9}
	AGD-SDAJ	67	70	1.90×10^{-9}
	Anderson-APM	103	109	3.08×10^{-9}
	SBB-Dual	200	212	3.01×10^{-9}
	Dykstra-APM	406	430	3.17×10^{-9}
cor3120 $n = 3120$	Newton-SIN-BH	6	6	2.33×10^{-9}
	AGD-SDAJ-BH	41	42	1.32×10^{-9}
	AGD-SDAJ	56	59	2.17×10^{-10}
	Anderson-APM	155	160	3.21×10^{-9}
	SBB-Dual	255	270	3.38×10^{-9}
	Dykstra-APM [‡]	–	300	1.16×10^{-5}
bccd16 $n = 3250$	Newton-SIN-BH	4	5	2.71×10^{-11}
	SBB-Dual	5	6	1.20×10^{-10}
	Anderson-APM	6	6	4.22×10^{-10}
	Dykstra-APM	6	7	1.46×10^{-10}
	AGD-SDAJ-BH	8	9	8.88×10^{-10}
	AGD-SDAJ [‡]	–	323	3.93×10^{-5}

8.1×10^{-12} , so forward errors on these matrices are resolved well below the targets used, and are insensitive to the starting point; the residual floor near 10^{-12} is reached within four to six Newton iterations. We therefore treat these runs as a check on external validity, not as measurements of the same quality as table 5. Every solver is run to a common tolerance of 10^{-9} , capped at 500 EVDS on the two smaller matrices and at 300 iterations on the two larger ones, and costs are read at the forward-error target 10^{-8} .

The matrices disagree with each other, broadly in the direction the synthetic family predicts. At $n = 94$, SBB-Dual needs 7 EVDS to 10^{-8} against 12 for AGD-SDAJ-BH and 11 for Dykstra-APM, which is the pattern of table 6 at $n = 100$, and Anderson-APM needs 6. At $n = 1399$, SBB-Dual costs three times AGD-SDAJ-BH (200 against 66), and at $n = 3120$ six times (255 against 41), which is the pattern of section 5.6. **bccd16** is easy for every method: all but one reach 10^{-8} within 8 EVDS, so it does not discriminate between the first-order methods. Newton-SIN-BH is cheapest on all four, as everywhere else in this paper. The change of order is therefore visible on matrices with uncontrolled spectra that we did not construct. With four matrices that differ in more than dimension, this is consistent with the dimension effect of section 5.6, not an independent

demonstration of it.

The line-search stall occurs on real data. On the Rocky Mountain matrix the naive-Armijo Newton control exhausts its budget at a forward error of 1.03×10^{-8} , while the same solver with the Borsdorf–Higham line search converges in 4 EVDs to 1.49×10^{-13} . On `cor1399` the naive control converges normally (13 EVDs against 7). On `bccd16` the same defect appears in unmodified AGD-SDAJ. Both AGD variants follow an identical path to a forward error of 3.93×10^{-5} at the fifth EVD. From there AGD-SDAJ-BH, whose safeguard accepts the step, converges in four more EVDs to 8.9×10^{-10} ; unmodified AGD-SDAJ rejects trial after trial in its monotone line search, with the error unchanged in its first eight digits, until it exhausts the budget at 323 EVDs. The defect of section 4.2 is thus not an artefact of the constructed family, and it affects both published methods on real data, but neither is it universal: whether it appears depends on the matrix. Unmodified AGD-SDAJ also stalls on 10 of the 81 data-seeded and 1 of the 20 perturbed instances of section 5.8, and on `mmb13`, `tyda99r2` and `bhwi01` (section B).

The cost of a reference. At $n \approx 3000$ the reference solve alone took 7 to 42 minutes in the recorded check run, and a first attempt at `bccd16` with an uncapped absolute tolerance ran for four hours without finishing. A reference accurate enough to measure forward error can cost more than the solves being measured. That is a practical argument for the synthetic family, whose solution is available in closed form at any size.

5.11 Accounting convention

Under all-trial accounting, every evaluated point counts toward reaching and holding a target, including rejected line-search trials. A rejected trial can then be credited with reaching a target, but one that lies above the target also breaks a hold. On this family the second effect dominates slightly. Median costs change only for the two AGD variants, and all-trial accounting is marginally the more expensive: for AGD-SDAJ-BH the median rises by 1.0, 1.5 and 2.0 EVDs at $\varepsilon = 10^{-4}, 10^{-8}, 10^{-10}$, and for unmodified AGD-SDAJ by 1.0, 2.0 and 2.0. Newton-SIN-BH, SBB-Dual and Dykstra-APM are unaffected, because they have few or no rejected trials here. The effect is negligible on this family and never changes an ordering. For methods with many rejected trials it need not be, and accepted-only accounting is the rule that cannot credit a point the method itself rejected.

Two further accounting choices deserve a check, because each could favour a particular method. First, cost is scored on the raw projection $\Pi_{\mathcal{S}_+^n}(G + \text{Diag}(y))$, while SBB-Dual applies a Borsdorf–Higham rescaling before returning its iterate (section 5.1), so it is scored on a quantity it does not return. Both columns are recorded. Rescoring every study on the rescaled iterate leaves the ordering unchanged at 19 of the 20 (dataset, target) pairs, and moves SBB-Dual’s median at $n = 500$ and $\varepsilon = 10^{-8}$ from 207.5 to 188 EVDs: the reported numbers are the conservative choice for SBB-Dual on cost, in the 19 pairs where the ordering is unaffected. The exception cuts the other way: on the data-seeded set at 10^{-10} , rescoring lowers SBB-Dual’s reach count from 81 to 45 instances, so the raw-projection score used throughout this paper is actually the *favourable* choice for SBB-Dual there (it lets more of its runs count as reaching the target), and the apparent change of order at that one (dataset, target) pair is a comparison over an easier subset rather than a genuine reversal. Second, the AGD line searches return their last trial if a backtracking limit of 60 is reached, and the method then moves to a point the trajectory records as rejected. This never occurs in the reported runs: no line search in any study exceeded 60 trials, and every AGD stall recorded here is a work-budget exhaustion, not a backtracking failure.

Table 14: Feasibility of the returned iterate at matched tolerance 10^{-11} , 270 instances. ‘diag err’ is $\|\text{diag}(X) - e\|_\infty$.

solver	median $\lambda_{\min}(X)$	worst $\lambda_{\min}$	median diag err	worst diag err
Newton-SIN-BH	-3.42×10^{-15}	-1.25×10^{-14}	8.38×10^{-15}	5.70×10^{-12}
AGD-SDAJ-BH	-3.40×10^{-15}	-1.36×10^{-14}	2.53×10^{-12}	7.20×10^{-12}
AGD-SDAJ	-3.40×10^{-15}	-1.39×10^{-14}	2.79×10^{-12}	4.01×10^{-8}
SBB-Dual	-3.42×10^{-15}	-1.49×10^{-14}	0	0
Dykstra-APM	-3.43×10^{-15}	-1.36×10^{-14}	1.49×10^{-12}	9.37×10^{-12}
Anderson-APM	-3.45×10^{-15}	-1.36×10^{-14}	2.12×10^{-12}	7.50×10^{-12}

The iteration caps are not uniform across methods, and no comparison in this paper relies on a capped run. At $n = 500$ the caps are 1200 iterations for SBB-Dual, 800 for Dykstra-APM and a 4000-EVD budget for the AGD variants, chosen so that each method’s own exit is reached on most instances within a fixed total runtime. SBB-Dual converged on all 270 instances (worst case 1075), so its numbers are uncensored, and the AGD-SDAJ medians are taken over the instances that reached each target, which understates the cost of the unmodified variant and therefore understates the benefit of the safeguard.

Dykstra-APM is the one method whose cap binds, on 90 of 270 instances, and it also had the smallest budget, so we re-ran those 90 at SBB-Dual’s cap of 1200 iterations. *All 90 reached the larger cap as well.* At 1200 iterations they end at a median forward error of about 10^{-6} : 71 of the 90 reach 10^{-6} (median 1099 EVDs, against 152 for SBB-Dual and 53.5 for AGD-SDAJ-BH at the same target), 6 reach 10^{-8} , and none reaches 10^{-10} . The censoring is therefore a property of the method on these instances and not of the budget it was given; a cap generous enough to uncensor Dykstra-APM would have to be an order of magnitude larger than any other method needs. We report its position as censored throughout.

5.12 What each solver returns

Methods also differ in what they return (table 14). All six return matrices with $\lambda_{\min}$ at roundoff level, so none is meaningfully more positive semidefinite than another. The diagonal is a different matter. SBB-Dual returns an exactly unit diagonal because of its rescaling epilogue, whereas every other variant returns a diagonal that is accurate only to its convergence tolerance. The worst-case diagonal error of unmodified AGD-SDAJ is 4.01×10^{-8} , about 3.6 orders of magnitude larger than that of any other variant, which is a direct consequence of the stalls in section 4.2. The rescaling epilogue has a cost of its own: on a nearly feasible input at loose tolerance, it can increase $\|X - X^*\|_F$. Borsdorf and Higham bound that increase by $\text{tol} \|X\|_F / (1 - \text{tol})$ [6, Lem. 3.2], which is small whenever the tolerance is tight but need not be at a loose one. A benchmark that reports only cost hides these differences. A user who needs an exactly unit diagonal and a user who needs minimal distance may not be best served by the same method.

Costs at each solver’s own exit when all are run to 10^{-11} , and the exit reasons behind them, are in the released analysis output; under that convention the ordering equals the tight-target ordering of section 5.3, because the common tolerance is tight.

Table 15: Solvers exiting when the stopping quantity specific to each method first reaches the common numerical threshold $10^{-7}n$ at $n = 100$ (for the dual methods this quantity is $\|\nabla\theta\|_2$; for Dykstra-APM and Anderson-APM it is the diagonal error of the PSD iterate), 270 instances, single-threaded timing.

solver	median EVDS	median time (s)	median $\ X - X^*\ _F$
Newton-SIN-BH	4	0.006	5.04×10^{-8}
Newton-SIN	7	0.009	5.04×10^{-8}
AGD-SDAJ-BH	20	0.024	9.58×10^{-6}
AGD-SDAJ	20	0.024	9.58×10^{-6}
SBB-Dual	18.5	0.020	7.73×10^{-6}
Dykstra-APM	40	0.045	1.93×10^{-5}
Anderson-APM	13	0.016	1.37×10^{-5}

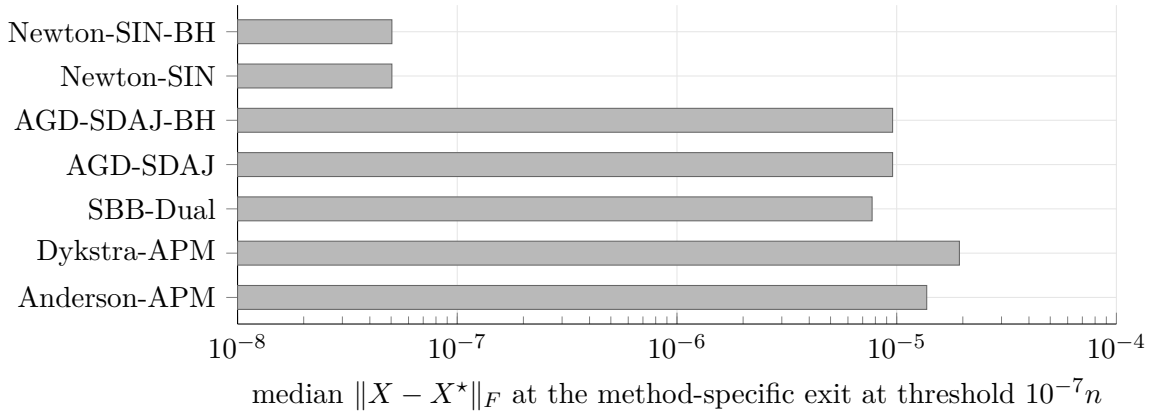


Figure 5: Achieved accuracy when each solver exits once its own stopping quantity reaches the threshold of the published dimension-scaled rule (tolerance 10^{-5} at $n = 100$; see table 15). Median forward errors range from 5.04×10^{-8} (Newton variants) to 1.93×10^{-5} (Dykstra-APM), a factor of 383. Costs measured at these exits are therefore costs at different accuracies.

6 The dimension-scaled stopping rule

A common exit criterion for dual NCM methods is $\|\nabla\theta(y_k)\|_2 \leq 10^{-7}n$. It is used, for example, in the experiments of [22]. Because the threshold grows linearly with n , the forward error it guarantees depends on both the dimension and the method. The general point that native stopping tests confound comparisons was made by Dolan, Moré and Munson [12]. This section documents a specific NCM instance of it.

Spread across methods at a fixed n . Table 15 and fig. 5 report the timing run under this rule at $n = 100$, where the tolerance is 10^{-5} .

At this dimension the best and worst median achieved accuracies differ by a factor of 383 over all 270 instances. Repeating the run at $n = 500$ on a subset of 18 instances (threshold 5×10^{-5}) gives a spread of only 13.8, with medians from 1.74×10^{-5} for Newton-SIN-BH to 2.41×10^{-4} for Dykstra-APM. The spread across solvers is therefore not itself a stable quantity, and we do not claim it as one; the next paragraph compares the two dimensions on the same instances. Newton overshoots the tolerance because a quadratically convergent step moves far past the threshold. First-order methods stop close to it, and the stopping quantity of Dykstra-APM and Anderson-

APM (the diagonal error of the PSD iterate) is not the dual gradient norm at all. The cost column is therefore a comparison at different accuracies, and it favours methods that stop early. At this tolerance the native costs also place SBB-Dual (18.5 EVDs) slightly ahead of both AGD variants (20 EVDs), with Anderson-APM ahead of all three (13 EVDs). This native order happens to coincide with the tight-target order of table 5, apart from the tie between the two AGD variants, but it is obtained at accuracies near 10^{-5} , which lie at the crossover of fig. 2 where the matched-accuracy order is changing; the agreement is therefore not evidence for either regime.

Growth with n . The rule loosens as n grows. In earlier runs, outside the ranking study, a nearly feasible $n = 500$ instance exited after *zero* updates with forward error 7.8×10^{-4} . At $n = 2000$, both dual first-order methods stopped after 3 to 5 iterations with forward errors between 2.7×10^{-3} and 4.1×10^{-3} . Under this rule, a larger problem can appear easier simply because less accuracy is being asked for. The paired family makes the same point with the dimension controlled. Each of the 18 instances run at $n = 500$ has a counterpart at $n = 100$ with the same rank, multiplicity, separation and seed label, so the comparison can be made instance by instance. The threshold loosens by a factor of five between the two. The first-order methods degrade steadily: the paired median ratio of the forward error at $n = 500$ to that at $n = 100$ is 10.1 for AGD-SDAJ-BH, 7.3 for SBB-Dual, 12.6 for Dykstra-APM and 9.9 for Anderson-APM. Per instance these ratios lie between 4 and 25 for the first three and between 3.8 and 42 for Anderson-APM. Newton-SIN-BH degrades by a paired median factor of 1.6×10^3 , but erratically, from 0.16 to 5.3×10^5 across the 18 instances. Its exit lands far inside the threshold, at a median gradient norm 8×10^{-4} times the threshold at $n = 100$ but 0.20 times it at $n = 500$: quadratic convergence moves in large discrete steps, so the margin by which it overshoots is itself erratic and collapses whenever the run exits one step earlier. The first-order methods stop at 0.45 to 0.99 of the threshold at both dimensions and so track it. On these same 18 paired instances the spread across solvers therefore narrows from 1653-fold at $n = 100$ to 13.8-fold at $n = 500$ (a matched, smaller-sample counterpart to the 383-fold spread reported over all 270 instances at $n = 100$ above): the rule does not degrade the methods equally, and it conceals most of the advantage of the method that converges fastest. A reader comparing published $n = 100$ and $n = 500$ results under this rule is comparing solutions of very different quality. The real-matrix runs of section 5.10 were made at a matched tolerance only, so they give no native-rule spread on real data.

Recommendation. Timing or iteration counts should never be reported without the achieved forward error (or, when X^* is unknown, a certified bound on it). Where a dimension-scaled rule is used for compatibility with earlier work, it should be accompanied by a run to a common tight tolerance.

7 Reproducibility

Independent implementations of the earlier solvers. SBB-Dual, semismooth Newton and Dykstra-APM were implemented independently in Python and in Julia, using *different* constructions of the reference solution. On all 14 shared instances, the two implementations agree on update counts and forward errors to every printed digit.

AGD-SDAJ: verified against the published counts. AGD-SDAJ exists only in Julia. Because this paper argues that implementation details can reverse conclusions, we did not report any result for our reimplementations on the KKT family until it had reproduced Huynh and Hwang’s

Table 16: AGD-SDAJ ($q = 2$): published counts (outer iterations / EVDs / nonmonotone backtracks / $\|G - X^*\|_F$) against our reimplementaion. Published values are from Table 3 and Table A1 of [22].

instance	published	this work	status
P9 bccd16, $n = 3250$	1 / 5 / 0 / 29.06	1 / 5 / 0 / 29.0563	exact
P8 cor3120, $n = 3120$	3 / 18 / 1 / 5.44	3 / 18 / 1 / 5.4437	exact
P7 cor1399, $n = 1399$	4 / 31 / 8 / 21.03	4 / 32 / 7 / 21.0339	one backtrack differs

own published counts on their own instances, two of three exactly and the third to within one backtracking decision. Those instances are three matrices from their test set, loaded under the same convention used elsewhere in this project (table 16).

P9 and P8 agree exactly. P7 agrees on outer iterations and residual norm, and differs by one backtracking decision: 7 rejected trials locally against 8 published. That single decision accounts for the whole difference in EVDs. Excluding backtracks, the local base cost is $32 - 7 = 25$ and the published base cost is $31 - 8 = 23$. The gap of 2 EVDs is exactly the early-termination saving described in the note to their Table 3: with the extra backtrack, their trajectory reaches the tolerance *at* the preconditioned point and skips the final gradient step, whereas ours reaches it one step later. We attribute the differing decision to floating-point differences. The reference implementation is in MATLAB and ours in Julia. The two call different LAPACK eigensolver drivers, and on a dense 1399×1399 problem, differences in the last bits of the spectrum are enough to flip a single acceptance test whose two sides are nearly equal. We did not search over parameters to make P7 match, because tuning to a target would be exactly the kind of hidden fitting that the protocol is designed to expose. Section B gives the full verification record, including three transcription defects found and corrected along the way.

Single-machine caveat. All timings come from one machine with single-threaded BLAS: Intel Core 7 150U (10 physical cores, 12 logical), 32 GB RAM, Windows 11 (build 26200), Julia 1.12.6 with OpenBLAS (ILP64) through libblastrampoline. Per-EVD ratios (table 3) may differ on other hardware and BLAS libraries, and with multithreading. Each configuration in table 15 (the native-rule timings) was timed once, after a discarded warm-up. The matched-tolerance measurement behind table 3 was repeated five times in total; section 3.6 reports the resulting run-to-run variance, which is large.

Artifact. The generator, the instance manifests with hashes, all solver code, the trajectory and timing CSV files, and the analysis scripts that produce every table in this paper are listed in section D. They are available at <https://github.com/Realosunboy6/nmc-exact-benchmark>. The three synthetic instance sets are archived in full, with a permanent identifier, at <https://doi.org/10.5281/zenodo.22967421>; the equity-derived sets are not redistributed (section D).

8 Discussion and limitations

What changed the conclusions, and what did not. On this family, none of the conventions we tested changed which method was cheapest. The accuracy target, the instance regime and the problem dimension did change the order below it. For SBB-Dual against AGD-SDAJ-BH, the changes of lower-place ordering with the accuracy target and with rank that we found at $n = 100$ recur at $n = 500$ only above $r = 50$ (sections 5.6 and 5.7); the pair that does not involve SBB-Dual, Dykstra-APM against AGD-SDAJ-BH, does not recur by $r = 150$ at $n = 500$. On matrices built

from equity returns the degree of invalidity reverses the order as well, both for SBB-Dual and for Anderson-APM against AGD-SDAJ-BH (section 5.8). The stopping rule changed the achieved accuracy by a factor of 383. The accounting rule (accepted-only against all-trial) changed cost ratios but no ordering. The cost unit (EVDs against wall-clock time) changed cost ratios and, in half of the ten timing sessions, the ordering below first place too: repeating the matched-tolerance timing measurement five times per dimension (section 3.6) shows the single cheapest method is unaffected in every session, but which method is dearest overall is not, at $n = 100$. Two line-search details changed whether methods converged at all. The summary that the evidence supports is conservative: the protocol matters for the *magnitude* of reported advantages, for the ordering of methods that are close in cost, and, for the cost unit specifically, sometimes for which method is worst; it did not matter for identifying a clear first place. Where methods are within a factor of about two, conclusions drawn under a single convention or a single timing session should be treated with caution.

Why an exact family. The regime dependence in section 5.5 is organized mainly by rank, one of the three quantities the construction controls. Finding it requires both an exact X^* , so that forward error is not contaminated by an approximate reference, and independent control of the spectral structure at the solution. Real matrices supply neither. The same machinery then showed the dimension dependence of that finding: at $n = 500$ the same rank effect, for SBB-Dual against AGD-SDAJ-BH, changes the order only between $r = 50$ and $r = 75$ (section 5.7), so a grid chosen at one dimension can miss the crossover at another.

Limitations.

- *Problem size.* Results are reported at $n = 100$ and $n = 500$ (270 instances each), with 135 more at $n = 500$ and $r \in \{75, 100, 150\}$ for every variant except unmodified AGD-SDAJ. The crossover between SBB-Dual and AGD-SDAJ-BH lies between $r = 5$ and $r = 20$ at $n = 100$ and between $r = 50$ and $r = 75$ at $n = 500$; on data-seeded instances at $n = 550$ SBB-Dual is still slightly behind at $r = 100$, and at $n = 2105$ (section 5.9) the gap closes to within one EVD by $r = 400$ without a resolved crossover. For this pair the crossover is therefore bracketed at $n = 100$ and $n = 500$ and only bounded below at the two larger dimensions, where it lies higher; the data do not support a functional form, and the largest panel contributes four instances per rank with weak replication. Without SBB-Dual, the exchange of Dykstra-APM and AGD-SDAJ-BH seen at $n = 100$ has not occurred by $r = 150$ at $n = 500$, and at $n = 2105$ the order of Anderson-APM and AGD-SDAJ-BH is not monotone in rank. Dykstra-APM is censored at $n = 500$ by its iteration cap.
- *Exact recovery of the instances.* The three synthetic sets (675 instances) are archived in full (section D), so a reader can use the exact instances of this paper. Regenerating them instead is reliable only on a LAPACK build that returns the degenerate kernel basis in the same form: another build may return it rotated, which yields different members of the same family; the recorded fingerprints detect this but cannot undo it. The equity-derived sets (sections 5.8 and 5.9) are a stricter case: built from vendor price data that may not be redistributed, they cannot be regenerated by a reader at all, from fingerprints or otherwise, and two of them (the perturbed $n = 2105$ panel and the bulk-level pilot of section 5.7) have no manifest or fingerprint recorded even for that limited check.
- *Synthetic family.* The KKT family is synthetic. Its G matrices are correlation-like but are not estimated from data, and all of them are close to feasible: $\lambda_{\min}(G)$ lies between -0.075

and -0.011 over the 540 paired instances at $n = 100$ and $n = 500$. The degree of invalidity, which reorders the methods on the equity-derived matrices, is therefore not a controlled factor of the family, and the findings that depend on it rest on perturbed matrices with computed references. Section 5.10 adds four matrices from the literature, from $n = 94$ to $n = 3250$, measured against a computed rather than an exact reference; for the two largest that reference is capped and reaches $\|\nabla\theta\|_2 \approx 10^{-12}$ rather than 10^{-13} , and no wall-clock times were taken. Four matrices cannot separate dimension from the other ways in which they differ. The equity-based instances of section 5.8 use a factor model of one correlation matrix as the exact solution; those instances are only mildly invalid and have three replicates per setting, while the perturbed set has five draws per σ and a computed reference. The sets of section 5.8 rest on one base correlation matrix and those of section 5.9 on a second, so there are two panels in all, not a sample of them.

- *Statistical uncertainty.* Aggregate medians at both dimensions carry cluster-bootstrap intervals (tables 5 and 7). Each dimension rests on five paired seeds per rank, that is 15 seed clusters, so inferential statements remain modest. Cell-level orderings rest on five instances each, and within a cell the adjacent pair AGD-SDAJ-BH against AGD-SDAJ is usually unresolved, so the count of distinct orderings should not be read as a count of real regime changes. We report counts and intervals rather than significance tests, because the reuse of each drawn U across all cells of its rank makes instance-level comparisons strongly dependent. The pooled medians at $n = 100$ weight the three ranks equally, so the target at which a pooled median changes sign depends on the rank grid, and no correction for multiple comparisons is applied to the cell-level and target-level comparisons, which are exploratory.
- *Single machine, five timings.* All timings come from one machine with single-threaded BLAS. Section 3.6 repeats the matched-tolerance measurement five times in all (the session reported in table 3 and four further sessions): a given solver’s per-EVD cost varies by a factor of 3.8 to 4.9 across sessions, depending on the solver, at least twice the $\leq 1.50\times$ spread between solvers within one session, so per-EVD ratios that differ by less than about 50% should not be read as ordered, and the specific rank of the middle four or five solvers, by either EVD count or wall-clock time, should not be treated as a fixed property of the methods (section 3.6). Only the leading position (Newton-SIN-BH cheapest overall) is robust across every session; Dykstra-APM is dearest overall in 8 of the 10 sessions across both dimensions, but not all.
- *Parameters.* Each method in the main study ran at one parameter setting: the published values for Newton-SIN-BH, AGD-SDAJ, Dykstra-APM and Anderson-APM (history length $m = 2$), and the specification of section E for SBB-Dual, whose step-size clamp is $\eta = 0.01$. As a check, we reran the 270 $n = 100$ instances against AGD-SDAJ-BH at $\eta \in \{0.001, 0.01, 0.05, 0.1, 0.2\}$ and at $m \in \{1, 2, 3, 5, 10\}$ (`code/sensitivity_eta_m.jl`). SBB-Dual’s median cost to 10^{-8} moves from 33.5 to 38.5 EVDs over two orders of magnitude in η , and the count of instances on which it is cheaper than AGD-SDAJ-BH stays at 165–167 of 270 throughout, so the SBB-Dual/AGD-SDAJ-BH comparison is not sensitive to this parameter. Anderson-APM is more sensitive: its median cost falls from 36 at $m = 1$ to 19–20 at $m = 5$ –10, and the count cheaper than AGD-SDAJ-BH rises from 210 to 257–261 of 270, non-monotonically past $m = 5$. The published default $m = 2$ (median 24, 240/270) is therefore not the most favourable setting for Anderson-APM against AGD-SDAJ-BH at $n = 100$ and 10^{-8} ; a larger m would only improve that comparison at those conditions, not weaken it, though this was not checked against the full 11 conventions of section 5.3 or at other dimensions. The sensitivity of the rank-dependent orderings of sections 5.5 to 5.9 to these

parameters was not checked.

- *Feasibility propositions.* Proposition 2.3 assumes uniform μ , and the extension to mixed μ is open (section A). Proposition 2.5 is stated for a only; no companion bound on b is given, since the released family lies in the fixed-rank regime where proposition 2.5 does not apply.
- *Scope of methods.* We benchmark six variants of five methods. Not included are the recent semismooth Newton method for general projection equations [2] and the factor-structured variant [7]. The main study runs Anderson acceleration only at the authors’ default history length $m = 2$; the *Parameters* item above reports a sensitivity check covering $m \leq 10$, but at $n = 100$ only.
- *Attribution of the line search.* Newton-SIN-BH uses the published Borsdorf–Higham globalization, but not the rest of their algorithm (section 5.1); adapting their near-equality safeguard to AGD-SDAJ, which gives AGD-SDAJ-BH, is ours, not the original authors’. AGD-SDAJ-BH is reported beside unmodified AGD-SDAJ on every set where both were run (all except the 135 higher-rank instances), and Newton-SIN-BH beside the naive-Armijo control Newton-SIN in sections 3 and 4.
- *The author’s own method.* SBB-Dual is from the author’s companion manuscript [23]. Two findings rest on it specifically, the bracketed positions of the rank crossover and its recurrence at a higher rank at $n = 500$; the others hold for pairs that exclude it (section 1).
- *Unweighted problem.* Only the unweighted Frobenius problem is considered.

Conclusion. An exact instance family with a prescribed solution structure makes it possible to separate stopping-rule, reference, accounting and primitive-work effects in NCM benchmarks. On 270 instances at $n = 100$, these effects changed cost ratios and the lower places of the ranking, but not the cheapest method. Repeating the study on 270 instances at $n = 500$ sharpened that finding: Newton-SIN-BH remains the leading method, and, for SBB-Dual against AGD-SDAJ-BH, the changes of lower-place ordering with the accuracy target and with rank recur, but only above $r = 50$, a rank the original grid did not reach; for Dykstra-APM against AGD-SDAJ-BH, the same rank-dependent reversal seen at $n = 100$ has not recurred by $r = 150$ at $n = 500$. Rankings below the leading method are therefore not stable properties of the solvers across the regimes tested; they depend materially on the benchmark regime, including its dimension, the rank of the solution and the size of the violation, and, for the specific rank at which a reversal occurs, on which pair of methods is compared. The one clear failure we found is a known finite-precision line-search defect, which appears in two separately published methods, on the constructed family and on real matrices. It is removed in the semismooth Newton method by the published Borsdorf–Higham modified line search, and in AGD-SDAJ by our adaptation of their near-equality safeguard. We recommend that NCM comparisons report forward error against a trusted solution at a common tolerance, give rejected trials no accuracy credit, and report primitive work and wall-clock time together.

A Proofs for the instance family

Throughout, $U \in \mathbb{R}^{n \times r}$ has rows u_i , $X^* = UU^\top$, P_r is the orthogonal projector onto $\text{range } U = \text{range } X^*$, and $Q_0 Q_0^\top = I - P_r$ (because X^* is symmetric, $\ker X^*$ is the orthogonal complement of $\text{range } X^*$).

A.1 Proof of proposition 2.3

With $\mu = ce$, we have $W = cQ_0Q_0^\top = c(I - P_r)$. For $i \neq j$, the diagonal correction $\text{Diag}(\text{diag } W)$ in (3) does not affect the entry (i, j) , and $I_{ij} = 0$, so

$$G_{ij} = X_{ij}^* - W_{ij} = X_{ij}^* + c(P_r)_{ij}.$$

By the triangle inequality, $|G_{ij}| \leq |X_{ij}^*| + c|(P_r)_{ij}| \leq a + cb$. If $a < 1$ and $c \leq (1 - a)/b$, then $a + cb \leq 1$. $\square$

Remark A.1. The bound is sufficient but not necessary. Cancellation between X_{ij}^* and $c(P_r)_{ij}$, and the fact that a and b are generally attained at different pairs (i, j) , both make it conservative. For this reason the generator checks $|G_{ij}| \leq 1$ entry by entry.

A.2 Proof of proposition 2.4

Fix $r \geq 2$ and $\epsilon \in (0, 1)$. Because S^{r-1} is compact, it can be covered by finitely many caps $C_1, \dots, C_N$, $N = N(\epsilon)$, each of angular radius $\epsilon/2$. Any two points in the same cap make an angle of at most ϵ , so their inner product is at least $\cos \epsilon \geq 1 - \epsilon^2/2$. Once $n > N$, the pigeonhole principle forces two of $u_1, \dots, u_n$ into a common cap, so $a \geq 1 - \epsilon^2/2$ deterministically. Since ϵ is arbitrary and $a \leq 1$ always, $a \rightarrow 1$ along every realization. (A probabilistic argument is needed only for a *rate*; the limit itself follows from pigeonhole.) $\square$

Remark A.2 (What this does not give). It is tempting to conclude that a fixed $c > 0$ must eventually violate proposition 2.3. That does not follow. At fixed r the projector bound b is observed to decrease with n as well (section 2.4), though we do not prove it, and if it tends to zero then $a + cb \leq 1$ can persist: with $a_n = 1 - 2/n$ and $b_n = 1/n$ one has $a_n \rightarrow 1$ while $a_n + cb_n < 1$ for every n at $c = 1$. A claim in either direction needs the relative rates of $1 - a$ and b , which we do not establish here.

A.3 Proof of proposition 2.5

Fix $i \neq j$ and condition on u_i . Since u_j is uniform on S^{r-1} and independent of u_i , we use the classical spherical-cap bound: for every unit vector v and every $t \geq 0$,

$$\Pr\{u_j^\top v \geq t\} \leq 2 \exp(-t^2 r/2).$$

This follows, with a factor of two to spare, from the cap estimate of [3, Lem. 2.2], which gives $\exp(-t^2 r/2)$ for $0 \leq t < 1$ (for $t \geq 1$ the probability is zero or trivially bounded); the subgaussian statement of [32, Thm. 3.4.5] gives the same decay with an unspecified absolute constant in the exponent, which would leave the constant in proposition 2.5 unspecified as well. Applying this with $v = u_i$, and again to $-u_j$, gives $\Pr\{|u_i^\top u_j| \geq t\} \leq 4 \exp(-t^2 r/2)$; the bound is unchanged after averaging over u_i . Taking a union bound over the fewer than $n^2/2$ unordered pairs,

$$\Pr\{a \geq t\} \leq 2n^2 \exp(-t^2 r/2).$$

Setting the right-hand side equal to η and solving for t gives $t = \sqrt{(2/r)(2 \log n + \log(2/\eta))}$, which is the stated bound. If $r/\log n \rightarrow \infty$ then $t \rightarrow 0$ for any fixed η . $\square$

Remark A.3 (On b). We do not state a companion bound for $b = \max_{i \neq j} |(P_r)_{ij}|$. Writing $P_r = VV^\top$ with V having orthonormal columns, $(P_r)_{ij}$ is an inner product of two rows of V , whose joint distribution is not that of two independent uniform vectors; a bound would need a model for

range U (for instance a Haar-random subspace) together with a justification that the row-normalized Gaussian U induces it. Since the released family lies in the fixed-rank regime of proposition 2.4, where proposition 2.5 does not apply, and since the generator screens $|G_{ij}| \leq 1$ entrywise in any case, we leave this open.

Remark A.4 (Scope). These propositions concern $|G_{ij}| \leq 1$ only. Indefiniteness of G ($\lambda_{\min}(G) < 0$) is not implied and is checked separately. The released family ($r \in \{5, 20, 50\}$, $n = 100$) is in the fixed-rank regime of proposition 2.4, not the high-dimensional regime of proposition 2.5.

A.4 Mixed μ

Write $k = n - r$ for the kernel dimension and e_j for the all-ones vector of length j (unrelated to the all-ones vector e used elsewhere for the unit-diagonal constraint). For the released $\mu = (\mu_{\text{bulk}} e_{k-m}, \delta e_m)$, write $Q_0 = [Q_b \ Q_\delta]$. Then $W = \mu_{\text{bulk}} Q_b Q_b^\top + \delta Q_\delta Q_\delta^\top = \mu_{\text{bulk}}(I - P_r) - (\mu_{\text{bulk}} - \delta) Q_\delta Q_\delta^\top$. The off-diagonal entries therefore pick up an extra term $(\mu_{\text{bulk}} - \delta)(Q_\delta Q_\delta^\top)_{ij}$, whose size depends on the choice of basis for $\ker X^*$ used to select Q_δ . A bound analogous to proposition 2.3 follows with b replaced by $b + \max_{i \neq j} |(Q_\delta Q_\delta^\top)_{ij}|$.

Remark A.5 (Open). We do not characterize $\max_{i \neq j} |(Q_\delta Q_\delta^\top)_{ij}|$. It depends on which orthonormal basis of $\ker X^*$ is chosen, and the generator fixes that basis through the eigendecomposition it computes, so the quantity is not determined by (n, r, m, δ) alone. A basis-free bound would need either a canonical choice of Q_δ or a bound uniform over admissible bases. We therefore state proposition 2.3 for uniform μ only, and the generator screens the mixed- μ instances entrywise rather than relying on a bound. This is why the released family is screened rather than certified.

B AGD-SDAJ verification record

Gate. Before running our reimplement of AGD-SDAJ [22] on the KKT family, we required it to reproduce the authors' own published counts on their own instances. An unverified reimplement that is then reported as losing would not be a defensible result.

Source and implementation. The source is Huynh and Hwang (2025), *Numer. Linear Algebra Appl.* 32(6), DOI 10.1002/nla.70045. We transcribed Algorithm 1 (AGD), Algorithm 3 (QN-SDAJ), Algorithm 4 (AGD-SDAJ) and the Section 4.3 parameters from the full text: $c = 10^{-4}$, $\rho = 0.5$, $\lambda_1 = \lambda_2 = 0.01$, $\epsilon_l = 10^{-5}$, $\epsilon_u = 10^8$, $q = 2$, $x_0 = 0$, the stopping rule $\|\nabla\theta\| \leq 10^{-7}n$, and $k_{\max} = 200$. The implementation is `agd_sda_j.jl` (standalone, used for verification) and `agd_sda_j_ncm` in `bench_sbb_dual.jl` (instrumented, used for the study). The instances are the Anymatrix CORRINV files already used elsewhere in this project, exported with the same loading convention.

Equation (6), as transcribed. The nonmonotone line search of Algorithm 3 is a Li–Fukushima-type rule [26] on the residual:

$$\|\nabla\theta(x + \alpha d)\|^2 \leq (1 + \epsilon_k) \|\nabla\theta(x)\|^2 - \lambda_1 \|\alpha \nabla\theta(x)\|^2 - \lambda_2 \|\alpha d\|^2, \quad \epsilon_k = \frac{1}{(k+1)^2}, \quad (6)$$

where, as implemented, k is the index of the inner step within its outer iteration (ϵ_i in Algorithm 2), so that $\epsilon_k \in \{1, \frac{1}{4}\}$ for $q = 2$. This indexing is our reading of the text, not stated explicitly there; with it, the published counts of table 17 are reproduced exactly on P8 and P9. The test uses $\lambda_1 = \lambda_2 = 0.01$, and $\alpha = \rho^j$ is backtracked from 1 over $j = 0, 1, 2, \dots$ (we write j for the

backtracking exponent, since m is the near-zero multiplicity of Definition 2.1). The merit function is $\frac{1}{2}\|\nabla\theta\|^2$. The quasi-Newton direction $d = -B^{-1}\nabla\theta$ need not be a descent direction for it, which is why a nonmonotone rule is needed. The paper also confirms that B_k is reset to the identity once x_k^{pre} is generated, and that Step 4 discards x_k^{pre} if it increases θ .

Before the full text was available, we tried two guesses: a Grippo–Lampariello–Lucidi rule on θ , and a Li–Fukushima rule. The second had the right general form but three wrong details: unsquared norms, a missing $(1 + \epsilon_k)$ slack, and λ_1, λ_2 attached to the wrong terms. The guess columns of table 17 were run before the acceleration-factor correction described below, so they are not directly comparable with the final column. On P8 their EVD counts bracketed the published one; on P7 both were well above it, which pointed to a remaining defect outside the line search (item 3 below).

Table 17: Outer iterations / EVDS / nonmonotone backtracks, and $\|G - X^*\|_F$, under the final transcription (eq6) and the two superseded guesses (run at an earlier code state; see text).

instance	published	eq6 (final)	GLL guess	LF guess
P9 bccd16	1/5/0, 29.06	1/5/0, 29.0563 (exact)	1/5/0	1/5/0
P8 cor3120	3/18/1, 5.44	3/18/1, 5.4437 (exact)	3/17/0	3/19/2
P7 cor1399	4/31/8, 21.03	4/32/7, 21.0339	6/40/3	5/48/17

Results. Under the final transcription, $\|G - X^*\|_F$ matched the published value to the displayed precision on all three instances. This showed that the objective, gradient, projection and stopping rule were correct, and that any discrepancy lay in the amount of work per iteration.

Three hazards when reimplementing Algorithm 4. Each of the following is easy to introduce when transcribing the method from its pseudocode. Each leaves the method converging to the correct solution while changing the work it does, so none is visible from final errors, only from work counts checked against the published ones. We record them for anyone reimplementing the method. All three occurred in our own first transcription and were found by exactly that check: the first two by accounting for the five published EVDS on P9 one at a time, the third because a gap in outer iterations on P7 remained once the line search was correct.

1. *Redundant EVD at an already-evaluated point.* The inner QN-SDAJ routine recomputed θ and $\nabla\theta$ at the point its caller had just evaluated. This is the same class of defect as section 4.1, and it was found in the same way. Fix: pass the caller’s values into the inner routine.
2. *Missing early termination at the preconditioned point.* The note to Table 3 of [22] states that the loop also stops whenever x_k^{pre} satisfies the stopping criterion, which saves at least one EVD (the trial at $\alpha = 1$), and two when $\beta \neq 1$. Without this check, the outer gradient step is always paid.
3. *Wrong acceleration factor in Algorithm 3.* The factor β_k^{pre} had been implemented with Algorithm 1’s $a_k = \|g\|^2$. That formula is the special case for $d = -\nabla\theta$. In Algorithm 3, $d = -B^{-1}g$, so the general form of their Section 2.3 (the acceleration of Andrei [1]) is needed:

$$\beta = \frac{-g^\top d}{(g_t - g)^\top d} \quad \text{if the denominator is positive, else } \beta = 1,$$

where g_t is the gradient at the accepted trial point, as in algorithm 2. This form reduces exactly to Algorithm 1's when $d = -g$, so it is consistent with both statements in the paper. *This defect caused the gap in outer iterations:* correcting it reduced P7 from 6 outer iterations to 4.

The cumulative effect on P7 was 46 EVDS $\rightarrow$ 40 (defects 1–2) $\rightarrow$ 43 (exact equation (6)) $\rightarrow$ 32 (correct acceleration factor), against a published 31.

The remaining P7 residual. See section 7. P7 matches in outer iterations (4) and residual norm (21.0339) and differs by one backtracking decision (7 against 8). That decision accounts for the whole difference in EVDS, through the early-termination saving. We attribute it to last-bit differences between the MATLAB and Julia eigensolvers. P8 and P9 reproduce every count exactly on the same code path. No further parameter search was carried out.

Conditions of use (all met in this paper). (i) The code and this record are published with the results. (ii) The P7 residual is disclosed. (iii) Published counts are printed beside local counts wherever the comparison appears (tables 16 and 17).

Checks against other published results. Beyond the AGD-SDAJ gate, every solver was run on invalid correlation matrices with published solutions or work counts (table 18). The matrices are those of the CORRINV collection [19], taken from its repository rather than retyped. On the ten small matrices (`high02`, `tec03`, `bhwi01`, `mmb13`, `fing97`, `beyu11`, `tyda99r1-r3`, `usgs13`), run at tolerance 10^{-10} , the five solvers of the original comparison return valid correlation matrices (Anderson-APM, added later, is checked in section C) whose distances $\|G - X\|_F$ agree to at least eight decimal places and whose matrices agree with Newton-SIN-BH's to within 2×10^{-10} , except unmodified AGD-SDAJ on `mmb13` (8.7×10^{-10}), which stops at its budget. The three large matrices (`cor1399`, `cor3120` and `bccd16`) were compared only at the published tolerances.

Table 18: Work counts reproduced from the literature. Dykstra-APM uses the stopping rule of the released code `nearcorr_new.m` of [20], tolerance n eps; run for the same number of iterations, our implementation returns the same iterate, bit for bit on nine matrices and to 1.4×10^{-15} on `mmb13`. Newton-SIN-BH is compared with Tables 6.5 and 6.6 of [5].

solver	source	matrix	published / ours
Dykstra-APM	[20], Table 1	<code>tec03</code> , $n = 4$	39 / 39 iterations
		<code>bhwi01</code> , $n = 5$	27 / 27
		<code>mmb13</code> , $n = 6$	801 / 801
		<code>fing97</code> , $n = 7$	33 / 34 ^a
	[20], Table 4	<code>usgs13</code> , $n = 94$	18 / 18
Newton-SIN-BH	[5]	<code>cor1399</code> , $10^{-7}n$	5 / 5 iterations
		<code>cor1399</code> , n eps	7 / 7; $\ G - X\ _F = 21.033911$ / 21.033911
		<code>cor3120</code> , $10^{-7}n$	4 / 4
		<code>cor3120</code> , n eps	6 / not reached ^b ; 5.443751 / 5.443751

^aAt iteration 33 the stopping quantity is 1.12 times the tolerance 7 eps, so the extra iteration is decided in the last bits. ^bThe residual floor near 10^{-12} lies above n eps = 6.9×10^{-13} with our eigensolver; the solution itself matches to all published digits.

On three of these matrices unmodified AGD-SDAJ shows the effect of section 4.2: on `mmb13` it stalls (budget of 200 000 EVDS exhausted, against 210 for AGD-SDAJ-BH), and on `tyda99r2` and `bhwi01` it converges, but slowly (549 against 35, and 66 against 21).

The Borsdorf–Higham adaptation. AGD-SDAJ-BH changes only the monotone Armijo test of Algorithm 1, line 5 (algorithm 2). When $|\theta_t - \theta| < \gamma u(1 + |\theta_t| + |\theta|)$ with $\gamma = 100$ and $u = 2^{-53}$, a trial that reduces $\|\nabla\theta\|$ is accepted. The nonmonotone rule (6) is unchanged. This adaptation is ours.

C Anderson-APM verification record

Source and implementation. Anderson-APM is a transcription of `nearcorr_aa.m` from the code released with Higham and Strabić [20] (github.com/higham/anderson-accel-ncm). It keeps the order of the convergence test and the update, the bookkeeping of the stored history, the modified Gram–Schmidt update of the QR factors, MATLAB’s `qrdelete` for the oldest column, the optional column dropping (off by default) and the returned matrix. Only the default least-squares option, QR with updating, is transcribed; the paper’s experiments use it.

What was compared. Their Tables 1, 3, 4 and 7 report iteration counts on matrices that are public: four small matrices at history lengths $m = 1, \dots, 6$ and three eigenvalue floors δ (72 counts), two fixed-element problems at $m = 1, \dots, 5$ (10 counts), and `cor1399` and `cor3120` at $m = 2$. The RiskMetrics matrices and the $n = 90$ matrix of their Table 4 are not public. Their stopping test is $\|Y - X\|_F / \|Y\|_F < n \text{ eps}$.

Table 19: Published Anderson-acceleration iteration counts reproduced exactly, by LAPACK symmetric eigensolver. `syevr` is Julia’s default and is used for every result in the paper; `syevd` differs from it only in rounding.

counts	published	exact, <code>syevr</code>	exact, <code>syevd</code>
small and fixed-element, excluding <code>mmb13</code>	64	57	61
<code>mmb13</code>	18	4	5
<code>cor1399</code> , <code>cor3120</code> ($m = 2$)	2	0	2
unaccelerated, <code>cor1399</code> , <code>cor3120</code>	2	2	2

Why counts differ. Under the default eigensolver the two large counts are 118 and 163, against published values of 124 and 174. Two checks separate an error of transcription from rounding at the stopping tolerance. First, an independent implementation that solves each least-squares problem from scratch, without updating, gives the same iterates as ours to within 2.3×10^{-12} in relative Frobenius norm until the stopping quantity reaches 10^{-10} , and the same iteration count at the tolerances 10^{-8} and 10^{-10} in all 82 small cases. Second, changing only the eigensolver moves the counts at $n \text{ eps}$ onto the published ones: under `syevd` both large counts are reproduced exactly, and 61 of the 64 small ones outside `mmb13`; the three that remain are one iteration apart. On `cor1399`, under the default eigensolver, the stopping quantity spends the four iterations before it crosses between one and four times the tolerance, so the step at which it first falls below is decided by the last bits of each eigendecomposition.

The matrix mmb13. This is the order-6 matrix on which unaccelerated alternating projections needs 801 iterations. Near convergence the stopping quantity stays within a factor of three of the tolerance for tens to hundreds of iterations, so the count at which it first falls below is decided by rounding, and the gaps are large, up to 56 iterations: at $\delta = 0$ and $m = 4$ the published count is 126 and ours 70. The two independent implementations disagree with each other by similar amounts on this matrix while agreeing on its iterates to 10^{-12} , and the published counts are themselves not monotone in m (117 at $m = 3$, 126 at $m = 4$). We regard the counts on this matrix as uninformative about correctness either way.

What is claimed. We claim agreement of the iteration, not of every published count: our iterates agree with an independent implementation, and our counts agree with the published ones wherever rounding at the stopping tolerance does not decide them. No result in this paper depends on those counts. The ranking study stops at $\|\text{diag}(X) - e\|_2 \leq 10^{-11}$ and scores the recorded trajectory, and it uses the default eigensolver, as every other solver does. The comparison scripts and their logs are listed in section D.

D Artifact manifest

The generator, the solvers, the analysis scripts, the instance manifests and every result file behind every table are available at <https://github.com/Realosunboy6/ncm-exact-benchmark>. Paths below are relative to the artifact root. The full synthetic instance sets (table 20) are archived separately, with a permanent identifier, at <https://doi.org/10.5281/zenodo.22967421> under CC-BY-4.0.

Instance binaries are not stored in the code repository, since they reach 1 GB at $n = 500$; the manifests there record a hash and four numerical invariants of every instance in every set except the perturbed equity panel at $n = 2105$ and the bulk-level pilot. The generator is seeded and reproduces those instances up to rounding: regenerated on the original machine, each G differs from the original by at most 10^{-17} in relative Frobenius norm and each X^* is bit-identical, although the hashes, taken on exact bits, differ. The script `code/check_instances.py` compares a regenerated set with the recorded invariants instead. On another LAPACK build the generator may receive a basis for the degenerate kernel of X^* in a rotated form, which yields different members of the same family; the check detects this, since a rotated basis moves the invariants by 15 to 20 per cent. A 12-instance subset of the original instances at $n = 100$ (2 MB) ships with the code repository; the full sets (675 instances, $n = 100$ and $n = 500$, about 1.6 GB) are on the Zenodo archive above.

Table 20: Artifact manifest: the files behind the instance family, the solvers and every table and figure in the paper.

file	contents
<i>Instances</i>	
<code>code/gen_degeneracy_family.py</code>	generator and screening (section 2)
<code>code/materialize_exact_ystars.py</code>	exact $y^* = -\text{diag}(W)$ for a suite
<code>data/canonical_n100/</code>	12 instances with G , X^* , y^* , manifest and cases
<code>data/manifests/</code>	SHA-256 manifests, numerical fingerprints and screening logs for the synthetic sets, the equity-derived sets with exact solutions and the perturbed panel at $n = 550$, and the four literature matrices

continued on the next page

Table 20: Artifact manifest (*continued*).

file	contents
code/check_instances.py	checks a regenerated set against the fingerprints
code/export_real_suite.py	literature matrices from Anymatrix CORRINV
code/export_thesis_kkt.py, export_thesis_suite.py	equity-derived sets (sections 5.8 and 5.9)
<i>Solvers</i>	
code/bench_sbb_dual.jl	sbb_dual, newton_ncm_globalized, agd_sdaj_ncm, dykstra_apm, trajectory recorder, study drivers
code/anderson_apm.jl	Anderson-APM, transcribed from Higham and Strabić's released code
code/agd_sdaj.jl	standalone AGD-SDAJ used for verification (section B)
code/validation/	checks against Higham (2002), Higham–Strabić (2016), Borsdorf (2007), Huynh–Hwang (2025), including the Anderson-APM record of section C; logs in results/validation/
code/export_agd_verify_matrices.jl	writes the three matrices used to check the AGD-SDAJ reimplementations (section B) from a local copy of Huynh and Hwang's own test matrices, not included
code/timing_rerun.sh	the driver behind the canonical (session 0) timing files in results/ (its own output files carry a _v2 suffix, renamed on shipping); repeat_timing.sh reuses its commands for the four further sessions
code/analyze_trajectory.py	an earlier prototype of analyze_ranking.py's cost-to-target logic, reading a trajectory.csv not shipped here; superseded, kept for provenance
code/highrank_queue.sh, anderson_rest_queue.sh	drivers for the runs added with Anderson-APM
code/check_reference_accuracy.jl	two differently initialized references per matrix (section 5.10)
code/sensitivity_eta_m.jl	the η/m sweep of section 8; output in results/analysis/sensitivity_eta_m.md
code/trace_armijo_stall.jl	the backtrack and affected-iteration counts of section 4.2; output in results/analysis/armijo_stall_trace.md
<i>Results and analysis</i>	
code/reproduce.sh	regenerates the analysis tables from results/*.csv.gz and the figure data, and prints which file holds which table; results/analysis/ also holds working files from earlier analysis passes that this script does not regenerate and that no table or number in the paper cites
results/ranking_kkt270_v2.csv.gz	one row per EVD, the five original solvers, 270 instances at $n = 100$
results/ranking_n500.csv.gz	the same at $n = 500$
results/ranking_highrank_n500.csv.gz	$r \in \{75, 100, 150\}$, Newton-SIN-BH, AGD-SDAJ-BH and SBB-Dual (table 8)

continued on the next page

Table 20: Artifact manifest (*continued*).

file	contents
results/ranking_highrank_proj.csv.gz	the same instances, Dykstra-APM and Anderson-APM
results/ranking_n500_dykstra1200.csv.gz	the 90 censored Dykstra-APM runs at $n = 500$ rerun at a cap of 1200 (section 5.11)
results/ranking_anderson*.csv.gz	Anderson-APM on every other test set, run after the original studies
results/ranking_thesis*.csv.gz	equity sets at $n = 550$ (tables 9 and 10)
results/ranking_us2105*.csv.gz	equity sets at $n = 2105$ (tables 11 and 12)
results/ranking_real.csv.gz	four literature matrices (table 13)
results/timing_kkt270_native.csv, timing_kkt270_matched.csv, timing_n500_native.csv, timing_n500_matched.csv	single-threaded timings of all seven variants, the session reported in tables 2, 3 and 15 (section 3.5)
results/timing_kkt270_matched_rep k .csv, timing_n500_matched_rep k .csv, $k = 1, \dots, 4$	four further matched-tolerance timing sessions, same instances and code, behind the session-to-session spread of section 3.6
results/timing_session1/	an earlier timing session, kept for comparison; not used in any table
code/analyze_ranking.py	tables 5 to 7, 10 and 14, the paired comparison of section 5.5 and the accounting numbers of section 5.11; output blocks 2, 3, 4, 5 and 7 of the script’s own markdown report (headed ## 2 etc., not this paper’s section numbers) use an earlier reach rule, kept for reproducibility and flagged in the output itself, and every number the paper cites comes instead from the corrected recomputation from output block 12 on, or, for table 14, from output block 6, which does not use that rule
code/analyze_by_rank.py	tables 8 and 9; results/analysis/us2105_by_rank.md is an earlier working copy of table 11, not regenerated by reproduce.sh and superseded by analyze_us2105.py’s output, below
code/analyze_per_instance.py	table 13
code/analyze_stopping_rule.py	the cross-dimension comparison of section 6
code/make_figure_data.py	the data behind every figure
code/analyze_highrank.py	supplementary seed-level diagnostics for section 5.7, in results/analysis/highrank_analysis.txt; neither table 8 nor table 12 itself carries a bootstrap interval column, and the cluster-bootstrap intervals quoted in the surrounding prose (e.g. the $r = 50$ crossover) come from analyze_ranking.py, above
code/analyze_us2105.py	tables 11 and 12, in results/analysis/us2105_analysis.txt, including the per-draw EVD pairs behind table 12’s entries; the $r = 400$ raw range quoted in section 5.9 (four instances, not independent seed clusters, so no cluster bootstrap applies) is this script’s own range column

continued on the next page

Table 20: Artifact manifest (*continued*).

file	contents
<code>code/analyze_confound.py</code>	the bulk-level pilot of section 5.7, from <code>results/ranking_confound_mu*.csv.gz</code>
<code>code/check_scoring_column.py</code>	the rescaled-iterate sensitivity check (section 5.11)
<code>code/analyze_timing.py</code>	tables 2, 3 and 15
<code>code/analyze_timing_repeats.py</code>	the five-session spread and the EVD-vs-wall-clock agreement check of section 3.6; output in <code>results/analysis/timing_repeats.md</code>
<code>code/repeat_timing.sh</code>	the driver that produced the four further timing sessions above
<code>results/analysis/us2105_kkt_cases.csv</code> , <code>us2105_pert_cases.csv</code>	per-instance descriptives of the two $n = 2105$ sets, copied from the <code>cases.csv</code> written by <code>export_thesis_kkt.py</code> and <code>export_thesis_suite.py</code> : the $\lambda_{\min}(G)$ range, the negative-eigenvalue counts and the clipped-entry counts quoted in section 5.9

Development notes. `docs/` keeps the session-by-session run notes behind this paper’s results, including the equity-export commands and the timing re-run protocol. They are a working log, not a specification: where a note cites a file that a later run superseded, the note says so rather than being edited to match.

Data not redistributed. The Anymatrix CORRINV matrices carry their own terms and are obtained from their own repository. The two equity panels are built from vendor price data that may not be redistributed; the exporters, the selection rules and all result files are included, the raw prices are not.

Figure data. The files `paper/figures/data/*.dat` behind every figure are written by `code/make_figure_data.py` from the result files above, and `reproduce.sh` regenerates them for comparison. The file `evd_vs_eps.dat` applies the cost-to-target rule of `analyze_ranking.py` (accepted-only, reach and hold) on a half-decade grid of ε ; at the five decade targets it reproduces table 5, up to that table’s rounding of half-integer medians. The data behind figs. 1 and 5, like tables 2, 3 and 15, come from the timing runs.

E Solver pseudocode

The solvers of section 5.1 as implemented; Newton-SIN is algorithm 1 with the naive test (5) in place of the near-equality branch and without the cached factorization. The iteration subscript k of section 5.1 is dropped throughout. Parameter values and the sources they are transcribed from are given there; the verification record for AGD-SDAJ is section B, and Anderson-APM, transcribed from the released code of its authors, is documented in section C.

Algorithm 1 Newton-SIN-BH (semismooth Newton, cached factorization, BH line search)

Require: G , tolerance τ ; $\sigma = 10^{-4}$, $\mu_{\text{BH}} = 0.9$, $\gamma = 100$, $u = 2^{-53}$

```

1:  $y \leftarrow 0$ ;  $(\theta, g, X, \Lambda, P) \leftarrow \text{EVAL}(y)$  ▷ keep the factorization
2: while  $\|g\| > \tau$  do
3:   Solve  $(V + 10^{-12}I)d = -g$  by CG, with  $V$  the generalized Jacobian from the cached  $(\Lambda, P)$ ;
   stop at  $\|r\| \leq \min\{0.1, \|g\|\}\|g\|$  or on negative curvature
4:   if  $d$  not finite or  $g^\top d \geq 0$  then  $d \leftarrow -g$ 
5:   end if
6:    $\alpha \leftarrow 1$ 
7:   loop
8:      $(\theta_t, g_t, X_t, \Lambda_t, P_t) \leftarrow \text{EVAL}(y + \alpha d)$ 
9:     if  $\theta_t \leq \theta + \sigma \alpha g^\top d$  then accept; break
10:    end if
11:    if  $|\theta_t - \theta| < \gamma u(1 + |\theta_t| + |\theta|)$  then ▷ BH near-equality branch
12:      if full step has  $\|g_t^{(\alpha=1)}\| \leq (1 - \mu_{\text{BH}})\|g\|$  then accept the full step; break
13:      end if
14:      accept the unit steepest-descent step  $y - g$  (one further EVD, unless  $d = -g$  already);
      break
15:    end if
16:     $\alpha \leftarrow \alpha/2$  ▷ at most 50 trials, else exit
17:  end loop
18:  update  $(y, \theta, g, X, \Lambda, P)$  from the accepted trial ▷ no recomputation
19: end while
20: return  $X$ 

```

Algorithm 2 AGD-SDAJ [22] (AGD-SDAJ-BH: with the marked branch)

Require: $G, \tau; q = 2, c = 10^{-4}, \rho = 0.5, \lambda_1 = \lambda_2 = 0.01, \epsilon_l = 10^{-5}, \epsilon_u = 10^8$; for BH, $\gamma = 100$,
 $u = 2^{-53}$

```

1:  $y \leftarrow 0; (\theta, g, X) \leftarrow \text{EVAL}(y)$ 
2: while  $\|g\| > \tau$  do
3:    $(y^p, \theta^p, g^p, X^p) \leftarrow (y, \theta, g, X); b \leftarrow e$   $\triangleright B_0 = I$ , reset every outer iteration
4:   for  $i = 0, \dots, q - 1$  while  $\|g^p\| > \tau$  do  $\triangleright$  QN-SDAJ inner steps
5:      $d \leftarrow -g^p \oslash b; \epsilon_i \leftarrow 1/(i + 1)^2; \alpha \leftarrow 1$ 
6:     backtrack  $\alpha \leftarrow \rho\alpha$  until, with  $g_t = \nabla\theta(y^p + \alpha d)$ ,
 $\|g_t\|^2 \leq (1 + \epsilon_i)\|g^p\|^2 - \lambda_1\alpha^2\|g^p\|^2 - \lambda_2\alpha^2\|d\|^2$ 
 $\triangleright$  eq. (6)
7:      $\beta \leftarrow a/b'$  with  $a = -g^{p\top}d, b' = (g_t - g^p)^\top d$  if  $b' > 0$ , else  $\beta \leftarrow 1$   $\triangleright$  general direction
8:      $s \leftarrow \beta\alpha d; y^+ \leftarrow y^p + s; (\theta^+, g^+, X^+) \leftarrow \text{EVAL}(y^+)$  (reuse the trial if  $\beta = 1$ )
9:      $b_\ell \leftarrow z_\ell/s_\ell$  if  $z_\ell/s_\ell \in [\epsilon_l, \epsilon_u]$ , else 1, for each entry  $\ell$ , where  $z = g^+ - g^p$ 
10:     $(y^p, \theta^p, g^p, X^p) \leftarrow (y^+, \theta^+, g^+, X^+)$ 
11:  end for
12:  if  $\theta^p > \theta$  then  $(y^p, \theta^p, g^p, X^p) \leftarrow (y, \theta, g, X)$ 
13:  end if
14:  if  $\|g^p\| \leq \tau$  then  $(y, X) \leftarrow (y^p, X^p)$ ; break  $\triangleright$  early exit at  $x^{\text{pre}}$ 
15:  end if
16:   $d \leftarrow -g^p; \alpha \leftarrow 1$ 
17:  loop  $\triangleright$  Algorithm 1, line 5
18:     $(\theta_t, g_t) \leftarrow \text{EVAL}(y^p + \alpha d)$ 
19:    if  $\theta_t \leq \theta^p + c\alpha g^{p\top}d$  then break
20:    end if
21:    if BH and  $|\theta_t - \theta^p| < \gamma u(1 + |\theta_t| + |\theta^p|)$  and  $\|g_t\| < \|g^p\|$  then break
22:    end if
23:     $\alpha \leftarrow \rho\alpha$ 
24:  end loop
25:   $\beta \leftarrow \|g^p\|^2/((g^p - g_t)^\top g^p)$  if the denominator is  $> 0$ , else 1
26:   $y \leftarrow y^p - \beta\alpha g^p; (\theta, g, X) \leftarrow \text{EVAL}(y)$  (reuse the trial if  $\beta = 1$ )
27: end while
28: return  $X$ 

```

Algorithm 3 SBB-Dual (clamped spectral gradient on the dual)

Require: $G, \tau, \eta = 0.01$

```
1:  $y \leftarrow 0; (\theta, g, X) \leftarrow \text{EVAL}(y)$ 
2: while  $\|g\| > \tau$  do
3:   if first update, or  $s^\top w \leq 0$ , where  $s = y - y_{\text{prev}}, w = g - g_{\text{prev}}$  then
4:      $t \leftarrow 1$ 
5:   else
6:      $t \leftarrow \min\{\max\{s^\top s / s^\top w, \eta\}, 2 - \eta\}$  ▷ clamped BB1
7:   end if
8:    $(y_{\text{prev}}, g_{\text{prev}}) \leftarrow (y, g); y \leftarrow y - tg; (\theta, g, X) \leftarrow \text{EVAL}(y)$ 
9: end while
10:  $X \leftarrow D^{-1/2} X D^{-1/2}, D = \text{Diag}(\text{diag } X)$ ; set  $\text{diag}(X) = e$  ▷ rescaling epilogue
11: return  $X$ 
```

Algorithm 4 Dykstra-APM (alternating projections with Dykstra correction)

Require: G, τ

```
1:  $X \leftarrow G; \Delta S \leftarrow 0; \Delta U \leftarrow 0$ 
2: loop
3:    $Y \leftarrow \Pi_{\mathcal{S}_+^n}(X + \Delta S)$  ▷ one EVD
4:   if  $\|\text{diag}(Y) - e\|_2 \leq \tau$  then return  $Y$  ▷ PSD half-iterate
5:   end if
6:    $\Delta S \leftarrow X + \Delta S - Y$ 
7:    $Z \leftarrow Y + \Delta U$  with  $\text{diag}(Z) \leftarrow e; \Delta U \leftarrow Y + \Delta U - Z; X \leftarrow Z$ 
8: end loop
```

Algorithm 5 Anderson-APM (alternating projections with Anderson acceleration, history m ; Higham and Strabić [20])

Require: G, τ , history length m (here $m = 2$)

```
1:  $Y \leftarrow G; \Delta S \leftarrow 0; z \leftarrow (\text{vec } Y, \text{vec } \Delta S)$ 
2: for  $k = 1, 2, \dots$  do
3:    $X \leftarrow \Pi_{\mathcal{S}_+^n}(Y - \Delta S)$  ▷ one EVD
4:    $\Delta S' \leftarrow X - (Y - \Delta S); Y' \leftarrow X$  with  $\text{diag}(Y') \leftarrow e$ 
5:    $g \leftarrow (\text{vec } Y', \text{vec } \Delta S'); f \leftarrow g - z$ 
6:   if  $\|\text{diag}(X) - e\|_2 \leq \tau$  then return  $X$  ▷ PSD half-iterate; the study's rule
7:   end if
8:    $m_k \leftarrow \min(m, k - 1)$ ; form  $\Delta F_k, \Delta G_k$  from the last  $m_k$  differences of  $f, g$ 
9:    $\omega \leftarrow \arg \min_{\omega} \|f - \Delta F_k \omega\|_2$  ▷ QR with updating
10:   $z \leftarrow g - \Delta G_k \omega$ ; unpack  $z$  into  $Y, \Delta S$ 
11: end for
```

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